

Nelvarx

STRUCTURAL ANALYSIS
OF COMPLEX SYSTEMS

- STATES
- RELATIONSHIPS
- LEGAL ACTIONS
- BOUNDED OUTCOMES

CONSOLIDATED DEMONSTRATION MANUAL

Structural Analysis of Chess Positions

Seven analytical blocks

From a reproducible example to a bounded mathematical specification

DOCUMENT	NLV-CHESS-CONSOLIDATED-EN
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STATUS	Public working draft for review

SCOPE AND BOUNDARIES

A bounded demonstration of structural analysis using specified chess positions, task criteria, and horizons. Internal checks are reported to the stated extent. This is not a claim of optimal chess play, universal validity, or independent external verification. Internal working materials are not included.

This release updates the consolidated English edition only; readiness of another language edition is not established here.

Overall Contents

The seven original analytical blocks are retained in sequence. All page numbers below, in the block contents, and in the footers refer to this consolidated file. Existing section numbers and bookmarks are preserved.

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READING AND REVIEW

Each block retains its own overview and detailed contents. Block 6 contains the active formal definitions; Section 7.6 records the clarifications already incorporated. The change log distinguishes those edits from unchanged analytical results.

Status: Public working draft for review. Internal checks and independent external verification are separate; the latter is not claimed.

Nelvarx

Material preserved.
Defensive availability differs.

LEGAL REPLIES

Base

0

A

1

B

0

APPLIED STRUCTURAL ANALYSIS

Structural Analysis of Chess Positions

Block 1. Problem Statement and a Reproducible Demonstration

Initial conditions, structural dependencies and outcome verification

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For review and independent reproduction. This document is not a complete chess analysis methodology or confirmation of its universality.

Document Overview and Contents

This block presents a self-contained demonstration of Nelvarx structural analysis applied to three bounded chess variations. It is intended for reading, substantive review and independent reproduction.

KEY RESULT

All three variations retain 32 pieces and pawns. After the queen reaches h4, White has 0, 1 and 0 legal replies, respectively.

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PUBLICATION BOUNDARY

The initial data, definitions, results and checks for this example are disclosed. Internal working tools and the complete analytical toolkit are not included.

EDITION STATUS

For substantive review and repeated reproduction. No independent external review has yet been recorded.

1.1 Core Principle

PRINCIPLE

Preserving a system's composition does not guarantee that a specified function remains available.

The analysis must establish not only which elements are present, but also the actions available to them, the constraints on those actions and the dependencies that determine the outcome. A change in an element's position can preserve its physical presence while removing a specific way of using it for the task.

This block examines the principle through a bounded chess example. It compares positions in which a particular defensive reply remains available with positions in which that reply is absent and the check cannot be answered.

Every material claim is accompanied by initial conditions, an observed change and a way to verify it. The output is a reproducible analysis, not merely a substantive interpretation.

1.2 Problem Statement

Object of analysis

The object of analysis is a sequence of chess positions arising from the standard starting arrangement after four player actions. The analysis considers the locations of pieces and pawns, the side to move, attack lines, obstructions and legal defensive replies.

The assessment is made from the perspective of White's ability to answer a specific check against its king. Black's actions are treated as opposition to that task.

Function under assessment

Remove check against the white king with one legal reply. This is a narrower task than winning the game, maintaining a good position or securing the king in the long term. Answering one check does not establish any of these broader outcomes.

Under the Laws of Chess, a move may not leave the player's own king in check. If the king is attacked and no legal reply exists, the position is checkmate. This is the criterion used to verify the outcome. [1]

Central question and required output

Which changes make a defensive reply unavailable while all pieces and pawns remain on the board? The analysis must explain the mating mechanism, distinguish conditions that matter to it from details that do not, and report the outcomes of controlled changes. Another reviewer must be able to reconstruct the positions and obtain the same results.

1.3 Working Definitions and Boundaries

Position state means the placement of every piece and pawn together with the side to move and the information required to determine legal actions. In this example, that information is reconstructed from the complete move sequence beginning at the standard starting position.

Structural relationship means a specific connection between elements of the position that is relevant to the function under assessment. The relationships considered here are attacks on squares, interposition on a line, obstruction of movement and the ability to remove a threat on the next move.

Element function means its verifiable contribution to the task being assessed. A pawn may, for example, preserve the ability to block a particular line of attack. That function belongs to a specific state and scenario; it is not permanently assigned to the pawn.

Function availability means that at least one legal action exists through which the contribution can be made at the required moment.

A critical condition for the outcome under assessment is a condition whose alteration changes that outcome in an explicitly specified comparison. Criticality must not automatically be extended to other positions.

The assessment is limited to the baseline sequence and two control variations. Overall opening evaluation, optimal strategy, the psychological causes of errors and the effects of time on the clock are outside this block’s scope.

The empirical component consists of reproducing positions and checking available actions in a system governed by fixed rules. It is neither a statistical study of many games nor evidence of the methodology’s effectiveness in other domains.

1.4 Initial Data

The analysis begins from the standard starting position. Each side has 16 pieces and pawns, and White moves first. [1] The sequence is used as an instructional example and is not attributed to a particular historical game.

1. f3 e5 2. g4 Qh4#

Q denotes the queen, + denotes check, and # denotes checkmate.

Action	Side	Movement
1	White	Pawn f2-f3
2	Black	Pawn e7-e5
3	White	Pawn g2-g4
4	Black	Queen d8-h4

Reproduction retains every piece on the board, not just those directly involved in the attack. Focusing on a few elements does not justify excluding the others in advance: any of them may potentially provide a defensive reply.

1.5 Reproducing and Checking the Example

The following public reproduction protocol applies only to this case. It enables the published positions and results to be checked; it does not constitute the complete working toolkit for structural analysis.

Item to check	Reproduction action	Expected output
Task definition	Fix the side, the function being assessed and its completion criterion.	An unambiguous task
States	Replay the moves from the starting position and check legality.	Reconstructed positions
Changes	Compare each move with the stated line, capability or constraint.	A checked change map
Defence	Check all legal ways to remove check in the final position.	The reply set, or its absence
Control variations	Reconstruct each altered sequence from the start.	A comparison of conditions and outcomes

Observation is distinguished from interpretation. Clearing f2 is a directly verifiable change in the position. Its significance for the later attack is checked against the h4-g3-f2-e1 line.

Each control variation is reconstructed from the standard starting position. It must not be created by arbitrarily relocating pieces without checking that the new state is reachable through legal moves.

1.6 Application to the Baseline Example

1.6.1. Changes in relationships

The route by which the queen reaches h4 differs from the line along which it subsequently attacks the king. A queen may move diagonally but may not pass through intervening pieces. [1]

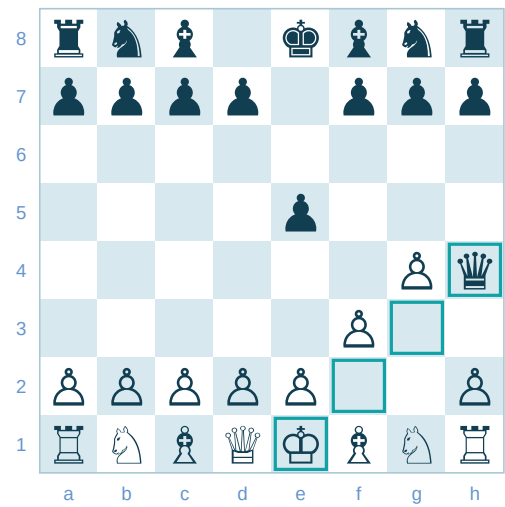
Move	Observed change	Significance for the mechanism
1. f3	The pawn clears f2.	The h4-g3-f2-e1 line to the king opens.
1... e5	The pawn clears e7.	The d8-e7-f6-g5-h4 route opens.
2. g4	The pawn leaves g2 and occupies g4.	The g2-g3 reply is removed.
2... Qh4#	The queen occupies h4.	Check is delivered with no legal defence.

Before g2-g4, the pawn was not on the line of the future check, but could occupy g3 on the following move and block it. After reaching g4, the pawn remains on the board but cannot return to g3: its movement rules do not permit a backward move. [1]

CHANGE	What is lost is not the pawn, but a specific legal defensive action by that pawn. Establishing checkmate also requires checking every other form of defence.
--------	---

1.6 Checking the Final Position

Position after 1. f3 e5 2. g4 Qh4#. White to move. All 32 pieces and pawns remain.



- Attack line: h4-g3-f2-e1.
- Intervening squares: g3 and f2.
- Side in check: White.
- Legal replies: none.

1.6.2. Checking king moves

The king is on e1. Every adjacent square within the board is examined:

Square	Occupancy	Why the king cannot move there
d1	White queen	A friendly piece obstructs the move.
d2	White pawn	A friendly pawn obstructs the move.
e2	White pawn	A friendly pawn obstructs the move.
f1	White bishop	A friendly piece obstructs the move.
f2	Empty	The square is attacked by the queen on h4.

Castling is prohibited while the king is in check; friendly pieces also obstruct it here. [1] **No legal king escape exists.**

1.6.3. Checking capture of the attacking queen

No white piece or pawn can legally capture the queen on h4 and remove check. In particular, the g4 pawn does not attack h4: it is an adjacent square on the same rank, not a forward-diagonal capture square. [1]

1.6.4. Checking interposition

Blocking the attack requires occupying g3 or f2. The pawn on f3 cannot return to f2, and the pawn on g4 cannot return to g3. No other white piece or pawn can legally interpose on this line.

1.6.5. Baseline result

The king is attacked and there are no legal replies. No captures or promotions occurred; the number, types and ownership of all elements remain unchanged. Material is preserved, but the defensive function is unavailable. This concerns White’s task, not the functioning of chess as a game.

1.7 Controlled Changes

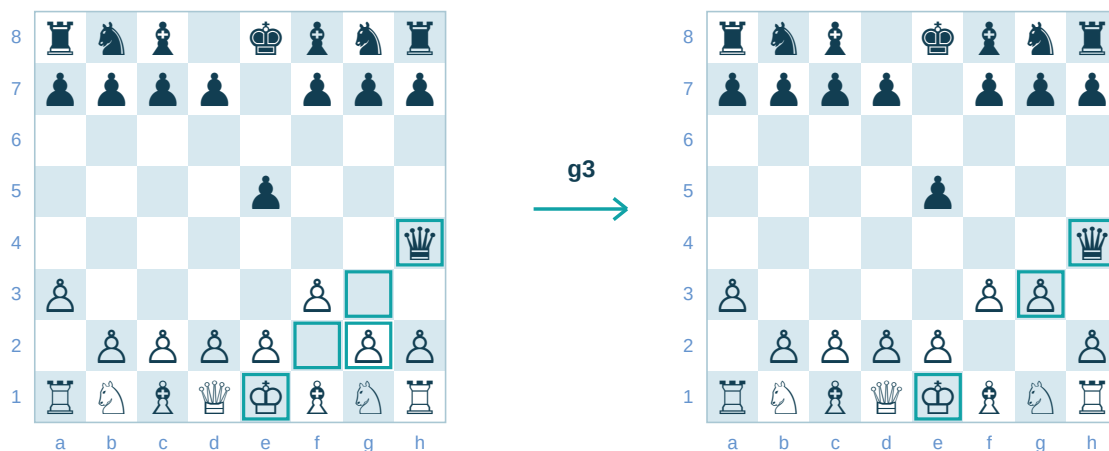
1.7.1. Test A. Preserving the defensive reply

Replace White's second move: play a2-a3 instead of g2-g4.

1. f3 e5 2. a3 Qh4+ 3. g3

The testable expectation is that leaving the pawn on g2 preserves the g2-g3 reply, blocking the attack. The test confirms it: the pawn occupies g3, the line between h4 and e1 is blocked, and check is removed. Immediately after Qh4+, this is White's only legal reply.

An open diagonal and the queen's arrival on h4 are insufficient for this checkmate. The absence of a blocking defence also matters. Replacing g4 with a3 changes the positions of two pawns relative to the baseline. It is a **controlled move substitution**, not a change to a single state variable. The significance of the g2 pawn is demonstrated by the specific g2-g3 action.



Left: after Qh4+, the only legal reply is g3. Right: after g3, the attack line is blocked. This does not establish the overall evaluation of the position or the outcome of the game.

1.7.2. Test B. Preserving the relevant function

Replace Black's first move: play e7-e6 instead of e7-e5.

1. f3 e6 2. g4 Qh4#

The testable expectation is that checkmate remains if clearing e7 matters, rather than the pawn's presence specifically on e5. The test confirms it: the queen's route is open, the attack line remains, and no legal defensive reply exists.

Two different locations of Black's pawn preserve the same relevant condition of this mechanism. **A change in an element's state does not necessarily change the outcome under assessment.**

1.8 Results Record

All variations are reproduced from the standard starting position. The comparison is made immediately after Black’s queen reaches h4.

Parameter	Baseline	Test A	Test B
Sequence	1. f3 e5 2. g4 Qh4	1. f3 e5 2. a3 Qh4	1. f3 e6 2. g4 Qh4
Pieces and pawns	32	32	32
White king in check	Yes	Yes	Yes
Legal replies	0	1	0
White's reply	None	g2-g3	None
Checkmate	Yes	No	Yes

The results were checked through position analysis and a repeated local enumeration of replies when preparing this edition. The verification is limited to the three final states and the g3 reply in variation A. It does not evaluate move strength or search for optimal play.

The defensive function is available in test A. In test B it is unavailable, as in the baseline. The technical reproduction record is provided in Appendix A.

1.9 Common Interpretation Errors

This register concerns risks in reading and using the case. It is not the methodology's internal classification system.

Risk	Incorrect interpretation	Correct verification boundary
R-01	The pawn's presence is treated as availability of its function.	Check the specific legal defensive move.
R-02	Losing g2-g3 is treated as sufficient proof of checkmate.	Exclude every other way of removing check.
R-03	The g2 pawn is assumed always critical and g4 always wrong.	Restrict the conclusion to the stated positions and task.
R-04	Removing check is treated as a favourable game outcome.	The g3 move establishes only that the current check is removed.
R-05	The selected sequence is treated as a forced victory.	Mate against every possible reply has not been established.
R-06	The example is treated as proof of universality or superiority.	Only a local dependency under specified conditions is checked.

ALEXEY GUDILIN

1.10 Acceptance Criteria and Outputs

The block may be accepted following repeated reproduction if the reviewer confirms the legality of every stated move, the absence of captures and the results in the record.

The key acceptance conditions are: no legal reply to check in the baseline or test B; g2-g3 is the only legal reply in test A; after that reply the white king is no longer in check.

Output	Content	Sections
Task specification	Object, function, boundaries and outcome criterion.	1.2-1.4
Change map	Each move's relationship to altered capabilities and constraints.	1.6.1
Verification record	Analysis of defence and comparison of control variations.	1.6-1.8
Analytical conclusion	The established dependency and its scope of applicability.	1.9-1.11

This edition records internal checks of the examples. Independent external verification is not claimed; further review and reproduction remain separate procedures.

1.11 Analytical Conclusion

The variations have the same material composition, but the availability of the defensive function differs. The difference arises not from the number of elements, but from their locations, movement rules, attack lines and available replies.

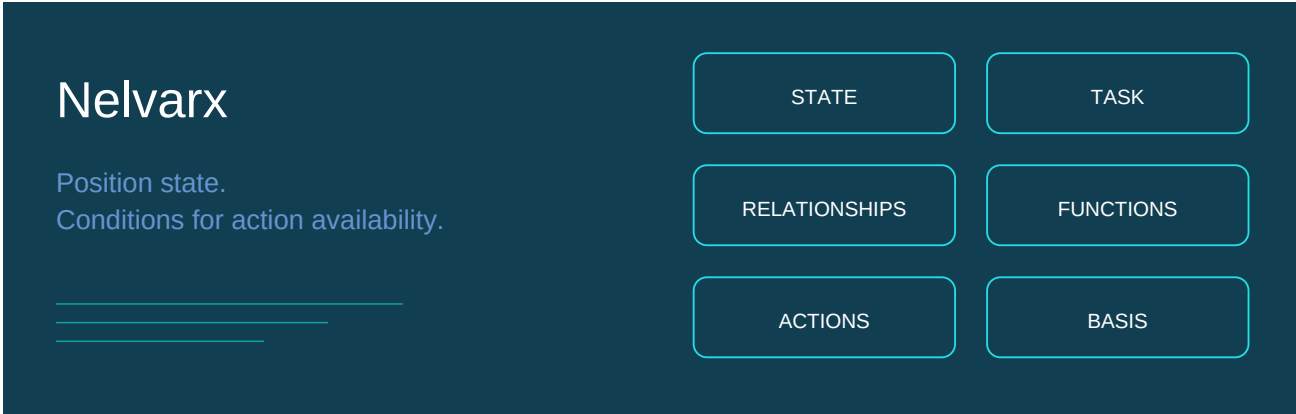
The first controlled change preserves an action that removes check. The second changes the location of Black's pawn while preserving the mating conditions. This distinguishes a relationship relevant to the outcome from a particular detail of an element's location.

BLOCK RESULT	From a record of movements to a testable dependency: a changed condition, a changed available action and a changed outcome.
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At this level, the methodology is not used to rename familiar chess events. It is used to define the task, identify relevant relationships and test an explanation through reproducible changes in conditions.

Status and next step

The block is available for review and independent reproduction. Mathematical formalization is not included in this block; the bounded specification is presented in Block 6.



APPLIED STRUCTURAL ANALYSIS

Structural Analysis of Chess Positions

Block 2. Structural Model of a Chess Position

States, relationships, functions and action availability

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Document Overview and Contents

Block 2 introduces a consistent structural model of a chess position and applies it to the variations from Block 1. It is intended for reading, substantive review and repeated reproduction.

OUTPUT	Position state, function, legal action and outcome are described separately and connected through verifiable relationships.
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PUBLICATION BOUNDARY

The definitions, domain-specific model, results and checks for the example are disclosed. Internal working tools and the complete analytical toolkit are not included.

EDITION STATUS

The model is applied to Block 1. Additional positions and extended checks are presented in Blocks 3–5; the bounded mathematical specification is presented in Block 6.

2.1 Core Principle and Purpose of the Block

PRINCIPLE

A structural model must show not only the arrangement of elements, but also the conditions under which their actions can perform a specified function.

Block 1 examined a specific dependency: the material composition of a position is preserved while the ability to answer check disappears. Block 2 introduces a consistent format for describing a position, needed for the subsequent analysis of further examples.

The purpose of the model is to connect the initial state, the analytical task, relationships between elements, legal actions and a verifiable outcome. These components must remain distinguishable. A piece's location is a property of the state. Its significance for defending the king is an analytical property that must be substantiated.

Output of the block: a structural description of a position that identifies the function being examined, the action through which it can be performed, and the conditions that make that action available or unavailable.

The model is not intended to determine the best move or provide an overall evaluation of positional strength on its own. At this stage, it establishes a basis for precisely describing and comparing the states under examination.

2.2 Source Basis and Scope of Analysis

The source material consists of three sequences from Block 1:

Baseline variation: 1. f3 e5 2. g4 Qh4#

Control variation A: 1. f3 e5 2. a3 Qh4+

Control variation B: 1. f3 e6 2. g4 Qh4#

The state of control variation A after the defensive reply **3. g3** is also examined.

All sequences are reconstructed from the standard starting position. They are used as instructional examples and are not attributed to specific historical games.

Move legality and the conditions for check, checkmate and stalemate are determined by the FIDE Laws of Chess, not by analytical interpretation. The terminology of the structural model is introduced in this manual to describe the dependencies under examination. ^[1]

The task from Block 1 is retained: **to establish whether check against White's king can be answered in a single legal reply.**

Relative to this task, Black is the opposing side. However, Black's pieces, available actions and the constraints they create are not excluded from the model. The full position is represented, not merely the defending side.

This block does not cover opening evaluation, long-term strategy, player psychology, clock time or predictions of a particular opponent's behavior. The sequence of moves is treated as an ordering of state changes, not as a measure of decision-making speed.

2.3 Components of the Structural Model

The model comprises six interrelated components.

Component	What is recorded	Question addressed
Position state	The full arrangement, the side to move and the information needed to determine legal actions	What is the initial state of the system?
Analytical task	The side being considered, the function under examination, the outcome criterion and the limit of the continuation	What exactly must be established?
Structural relationships	Attacks, obstructions, lines of interaction and other relationships relevant to the task	Which relationships determine the possibilities under examination?
Functions of elements	The verifiable contribution of individual pieces and pawns to performing the task	Which element and which action can produce the outcome?
Legal actions	Moves available to the side to move and the states that result from them	What can actually be done now?
Basis for the conclusion	Verified continuations, excluded replies and limitations of the analysis	What supports the conclusion?

These components must not substitute for one another. The presence of a pawn on a particular square does not prove that it performs a defensive function. Specifying a goal does not establish its attainability. The presence of an attack line does not prove the absence of a defense.

Relationships, legal actions and functional conclusions must correspond to the same reconstructed position. Piece locations must not be taken from one variation, the side to move from another and a defensive possibility from a third.

The state and the analytical task must also be recorded separately. Changing the research question does not move the pieces on the board, but it may change which relationships need to be regarded as significant.

2.4 Describing the State and Defining the Task

2.4.1. Position State

The definition of state from Block 1 is retained: the locations of all pieces and pawns, together with the side to move and the information needed to determine legal actions.

For each position, the occupancy of every square must be reconstructible. Either a complete record of the arrangement or an unambiguous move sequence from a known initial state may be used.

Selecting particular pieces for the explanation does not exclude the others from the model. A piece that does not participate directly in the attack under examination may obstruct a king move or provide an alternative defense.

The arrangement alone is insufficient: castling rights and the availability of an en passant capture depend on prior actions. ^[1] In this case, the necessary information is reconstructed from the complete move sequence.

For future tasks involving repetition of positions or other conditions for ending a game, the relevant history will need to be included in the description. Without supplementation, the current model of the immediate defensive reply must not be presented as a model of every possible basis for ending a game.

If a significant part of the state is unknown, it must be marked as undetermined. Missing information must not be silently replaced with a convenient assumption.

2.4.2. Analytical Task

The task is defined by the side, the function, the completion criterion and the analysis horizon.

Side under consideration: White.

Function under examination: to answer the existing check against White's king.

Completion criterion: after a legal reply, White's king is not in check.

Analysis horizon: one reply by White after the specified check.

Here, one move means a single action by one player. Black's subsequent play is not included in the demonstration that this narrow task has been performed, but it may require the analysis to be continued separately.

Answering the current check does not mean winning the game, preserving all pieces or ensuring the king's long-term safety. Those outcomes require different criteria and a different horizon.

Task completion must be assessed only against the outcome defined in advance. The analyst must not first verify that one check can be answered and then present that conclusion as confirmation of the position's overall resilience.

2.5 Structural Relationships

A structural relationship is a specified connection between elements of a position that is significant for the function under examination.

The model records the type of relationship, the elements involved and the state in which the relationship exists. Stating that pieces are connected is insufficient: the nature of the connection must be established.

Relationship type	Operational meaning	Example from the case
Square occupancy	An element occupies a particular square and affects how that square can be used	In variation A, a white pawn occupies g2
Line of interaction	A sequence of squares between elements is specified	g3 and f2 lie between the queen on h4 and the king on e1
Attack on a square	A square is attacked by a particular piece under the rules of the game	After Qh4+, the queen attacks e1 along the open diagonal
Obstruction to movement	An element or the state of a square excludes the movement under examination	The white bishop on f1 prevents its own king from moving from e1 to f1
Blocking a line	Occupancy of an intermediate square interrupts the line-based action under examination	After 3. g3, the pawn on g3 blocks the diagonal h4–g3–f2–e1
Condition for performing a function	An action's availability depends on a specified combination of circumstances	The reply g2–g3 requires a pawn on g2, an empty g3 and a legal resulting state

The final row describes not a single spatial adjacency, but an action's dependence on several conditions. That dependency must be checked as a whole.

Distinguishing an Attack from a Legal Move

Attacking a square is not equivalent to being able to make a legal move to it. Under the FIDE Laws of Chess, a piece may be considered to attack a square even if moving there would expose its own king. ^[1]

An attack map therefore cannot be obtained simply by combining legal moves. Likewise, the ability to move a pawn to an empty square does not mean that the pawn attacks that square.

In control variation A, the pawn on g2 can move to g3, but its attacked squares are f3 and h3. The defensive significance of g3 lies in blocking the line, not in attacking the queen.

Distinguishing an Existing Relationship from a Possible One

Before 3. g3 is played, the line h4–g3–f2–e1 is open. After the move, it is blocked. The possibility of blocking and an already established block belong to different states.

The record must therefore not state that the pawn on g2 already closes the diagonal. The correct formulation is: **from this position, the pawn can create a block by making a legal move.**

In this block, the space of relationships means the set of these specified connections and dependencies. It is not an additional object independent of the position, but a structural description of that position relative to the defined task.

2.6 An Element's Function and the Availability of an Action

2.6.1. Function as a Verifiable Contribution

An element's function is defined by its contribution to a specific outcome, not by the generic name of the piece.

For the pawn on g2 in control variation A, the record takes the following form:

Element: white pawn on g2.

Action: the move g2–g3.

Structural change: occupancy of an intermediate square on the attack line.

Outcome: the check against the king on e1 is answered.

The function is not assigned to the pawn unconditionally. It applies to this position, this side to move and this analytical task. After the state changes, the same element may acquire a different role or lose the ability to perform its previous action.

2.6.2. Three Distinct Claims about an Action

When describing an action, the following must be distinguished:

Compliance with the movement pattern. The movement under consideration complies with the piece type and the conditions for passing through or occupying squares.

Move legality. The action satisfies all applicable constraints of the position. In particular, it cannot leave its own king in check. ^[1]

Performance of the function under examination. The legal action achieves the outcome specified in the task definition.

For the current task, these checks are especially closely related: every legal reply by a side in check must answer that check. For other tasks, the legality and usefulness of an action may diverge. The model therefore retains them as distinct properties.

2.6.3. Function Availability Statuses

Four statuses are used in the description.

Available. In a state where the task applies, at least one legal way of performing it has been established.

Unavailable. The task applies, and examination of the complete set of legal actions confirms that there is no way to perform it.

Not required in the current state. The condition that makes the action necessary is absent. For example, the existing check has already been answered.

Undetermined. The initial data or the verification performed is insufficient for a conclusion.

Failure to find a reply is not equivalent to proving that no reply exists. The status unavailable requires a completed check, not merely an unsuccessful attempt to find a defense.

A defensive possibility after a hypothetical opponent move must be described separately from the actions available now. A White move must not be included in the current set of actions when it is Black's turn.

2.7 Completed Model of the Control Position

2.7.1. State Record

Identifier: control variation A, the state after 2... Qh4+.

Reconstruction sequence: 1. f3 e5 2. a3 Qh4+

Relative to the starting position, four elements have changed location: White's f2 pawn is on f3, White's a2 pawn is on a3, Black's e7 pawn is on e5, and Black's queen has moved from d8 to h4. All other pieces and pawns retain their original squares.

Side to move: White, move three.

Material composition: 32 pieces and pawns; no captures or promotions have occurred.

Significant squares: White's king is on e1; Black's queen is on h4; a white pawn is on g2; g3 and f2 are empty.

Analytical task: to answer check against White's king with one legal reply.

The history preserves castling rights because the kings and rooks have not moved. However, holding a right does not mean that castling is available in this position: White's king is in check and the paths are occupied. ^[1]

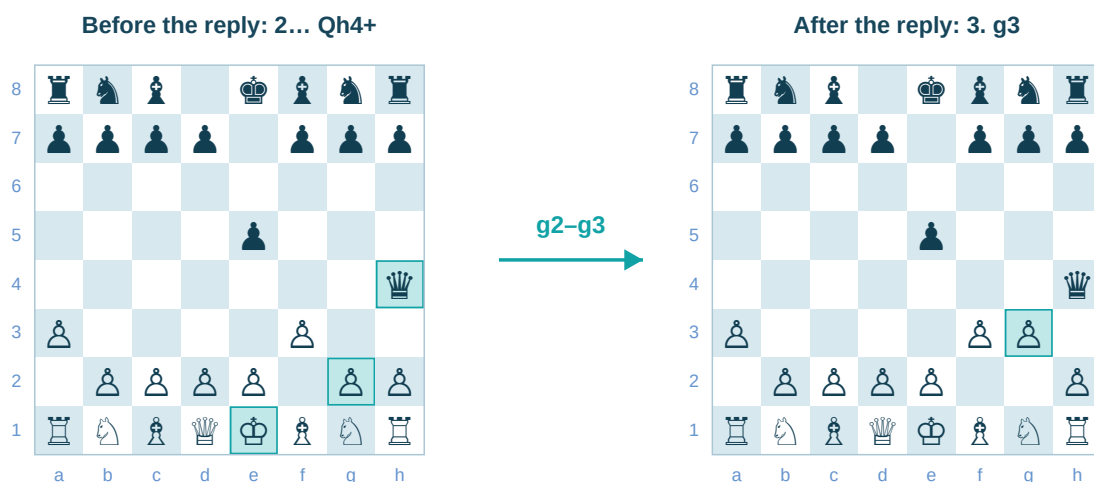
No en passant capture is available after the last queen move. The reconstructed history therefore adds no hidden defensive reply.

2.7.2. Significant Relationships

The queen on h4 attacks the king on e1 through the empty squares g3 and f2. The pawn on g2 does not block the line, but can occupy g3 on the next move.

For king escapes, d1, d2, e2 and f1 are occupied by friendly elements. The square f2 is empty, but attacked by the queen. These relationships are included in the model even though the relevant white pieces did not move in the sequence under examination.

The initial relationship map therefore includes not only the attacking piece and the potential defender, but also the constraints on alternative defensive methods.



Control variation A. Left: White to move and an open attack line. Right: the block on g3 and Black to move. All 32 pieces and pawns are shown.

2.7 Control Position: Actions and Outcome

2.7.3. Checking Specific Actions

Action considered	Complies with the element's movement pattern	Legal in this position	Significance for the task
g2–g3	Yes	Yes	Blocks the diagonal and answers check
g2–g4	Yes	No	Leaves the intermediate square g3 empty and does not answer check
h2–h3	Yes	No	Does not change the attack line to the king

The table distinguishes a pawn's movement pattern from the legality of a particular move. The actions g2–g4 and h2–h3 do not become defensive replies merely because they follow the usual direction of pawn movement.

The square g3 is attacked by Black's queen, yet the pawn's move there is legal. The prohibition on moving to an attacked square concerns the king's safety and is not a general prohibition for all pieces. ^[1]

The three rows illustrate the distinctions. The claim that the defense is unique rests not on selecting these three actions, but on examining the complete position and every way of answering check.

2.7.4. Resulting State

The complete set of White's legal replies in the control position contains one action:

g2–g3.

After it is played, the white pawn occupies g3, the attack line is blocked, and White's king is no longer in check. The turn passes to Black.

For the initial position, the function has the status **available**. For the resulting state, the immediate task of answering the existing check has the status **not required**, because the check has already been answered.

This distinction prevents a recording error in which the absence of a need for a defensive reply is interpreted as a loss of defensive capability.

2.8 Comparing States with a Single Model

The same descriptive fields are applied to the baseline variation, the two control variations and the state after the successful reply.

Parameter	Baseline after Qh4#	Variation A after Qh4+	Variation B after Qh4#	Variation A after 3. g3
Side to move	White	White	White	Black
Material composition	32	32	32	32
White's king in check	Yes	Yes	Yes	No
White's legal reply to the current check	None	g2–g3	None	Not required
Status of the function under examination	Unavailable	Available	Unavailable	Not required in the current state
Established outcome	Checkmate	Check with a unique reply	Checkmate	The original check is answered

During preparation of this block, controlled reproduction and auxiliary enumeration of ordinary moves confirmed **0, 1 and 0 legal replies**, respectively, for the three positions in check. This check applies to the specified states and does not constitute an independent assessment of the methodology as a whole.

The comparison yields three results.

The same material composition can coexist with different function availability. The number of elements does not determine whether a defensive reply exists.

Different states can yield the same result under the selected criterion. The black pawn's location differs between the baseline and variation B, but the absence of a defense is preserved. This does not make the positions identical in every other respect.

A successful action changes whether the task still applies. After g3, the initial task has been completed. Subsequent analysis must begin from the new state, the new side to move and an explicitly formulated question.

The model thus allows comparison not only of element locations, but also of the status of the function under examination.

2.9 Common Model-Building Errors

Error	Consequence	Required correction
Only part of the board is described	A defensive resource or obstruction is missed	Reconstruct the complete position; shorten only the explanation
An attack on a square is equated with a legal move	Constraints and king safety are determined incorrectly	Separate attack relationships from the set of legal actions
Different moments in the sequence are mixed	A future possibility is presented as an existing relationship	Specify the state for each relationship and action
No outcome criterion is defined	Answering check is replaced by an overall positional evaluation	Fix the function and horizon before comparing variations
Uncertainty is labeled as unavailability	An incomplete check becomes a categorical conclusion	Use the status undetermined until the check is complete
A checkmate symbol in the notation is treated as proof	The initial annotation replaces analysis	Reconstruct the position and check the replies independently

The completeness of negative claims must be checked separately. To confirm that a defense exists, it is sufficient to exhibit one legal reply. To confirm its absence, every legal way of performing the function under examination must be excluded.

The number of available moves must not be used as a standalone measure of positional quality either. In this case, it answers the narrow question of whether a defensive reply exists; it does not measure overall playing strength.

2.10 Acceptance Criteria and Outputs

The model is accepted for further use when the initial description is complete, actions are reproducible and the conclusion is traceable.

Completeness

Another reviewer must be able to reconstruct the full position and the side to move without additional assumptions. The function under examination, the outcome criterion and the horizon must be stated explicitly.

Reproducibility

For every claimed way of performing the function, the specific action and the resulting state must be provided. A claim that no such way exists requires a completed check of legal replies.

For the control examples, the description must reproduce the results in the table in Section 2.8. After the reply g3, it must not retain an outdated record of check or of the previous side to move.

Traceability

Every functional conclusion must be connected to a state, an action and a change in relationships. The statement that a pawn provides a defense must be explained through a specific move and its outcome.

Outputs of the block: a position record, a description of significant relationships, an element-function record expressed through an action and an outcome, a state-comparison table and a list of limitations.

These materials form a domain-specific description of the chess example. Their readiness does not mean that the model has already been tested on other types of position.

2.11 Requirements for Further Development

Block 3 applies the model to positions in which several functions depend on one element, or in which an action's legality does not coincide with its suitability for the selected task.

For such examples, the distinctions introduced between state, relationship, action and outcome must be retained. A new concept may be introduced only when the existing description is insufficient to represent the dependency under examination precisely.

Block 6 defines the relevant sets of states and actions, transition rules, function criteria, and limits of the continuations to be checked. The formalization must reproduce the domain-specific results rather than replace them with general notation.

The model is described here and applied to the examples from Block 1. Its further applications, extended checks, and bounded mathematical formalization are included in Blocks 3–6.

2.12 Analytical Conclusion

Block 2 introduces a consistent model connecting the full position state with the function under examination, significant relationships, legal actions and the basis for the conclusion.

Its application distinguishes the presence of an element from the availability of its function, a possible action from a legal move, and an existing relationship from one that arises after a transition. The distinction between an absent defense and the absence of a need to answer a threat that has already been resolved is recorded separately.

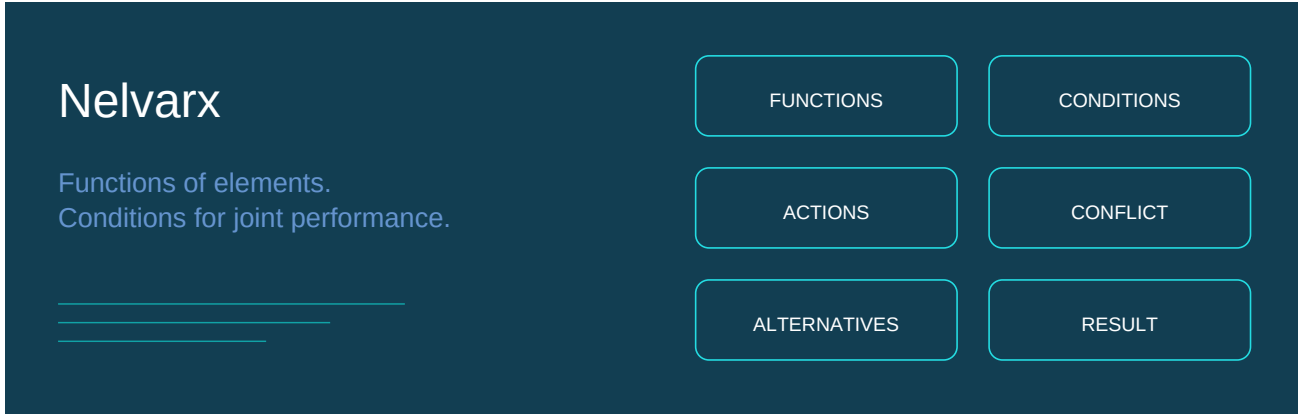
OUTPUT	The main result of this block: an element's functional significance is described not in isolation, but through a specific action, the conditions of its legality and a verifiable change of state.
--------	---

On this basis, the next block examines the functions and dependencies of elements in positions involving several interacting tasks.

Public Rules Basis

[1] FIDE. Laws of Chess. The rules source cited in the main text. Basis for move legality and the conditions for check, checkmate and stalemate.

<https://handbook.fide.com/chapter/E012023>



APPLIED STRUCTURAL ANALYSIS

Structural Analysis of Chess Positions

Block 3. Functions and Dependencies of Elements

Compatibility of functions, constraints, and alternative performing elements

DOCUMENT	NLV-CHESS-B03-EN
VERSION	0.2
AUTHOR	Alexey Gudilin
ISSUE DATE	22 September 2026
STATUS	Public working draft for review

For reading and repeated reproduction. This is not a complete methodology for chess analysis or evidence of its universality.

Document Profile and Contents

Block 3 applies the structural model to functions and dependencies of elements. It examines restrictions on a knight, compatibility of a capture with queen preservation, and alternative ways to perform the task.

RESULT	A restriction on an individual piece and the absence of a solution for the entire side are checked separately. A legal move does not necessarily meet every task requirement.
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BOUNDARY OF THE PUBLIC MATERIAL

The document presents domain definitions, instructional positions, results, and checks. Internal working tools and the complete methodological toolkit are not included.

DRAFT STATUS

Version 0.2 is a Public working draft for review. Further domain analyses are presented in Blocks 4–5, the bounded model in Block 6, and the final review in Block 7.

3.1 Core Principle and Purpose of the Block

CORE PRINCIPLE

Assigning several functions to one element does not establish that they can be performed together. Compatibility depends on specific actions, the conditions under which they are legal, and their consequences for the other functions.

Block 1 established the distinction between preserving the material composition of a position and retaining an available defensive response. Block 2 introduced a structural model linking the state, task, relationships, functions of elements, and legal actions.

Block 3 applies this model to dependencies between functions. It examines situations in which a piece participates in several relationships at once but cannot freely use every resulting possibility. It also tests whether another element can perform the required action without violating the condition that must be preserved.

Central question: when does a restriction on an individual piece make its assigned action impossible, and when can a different allocation of functions compensate for that restriction?

The block should produce verified descriptions of three mechanisms: an action prohibited by king safety, a legal action that conflicts with the selected outcome criterion, and a capability restored through another element.

These mechanisms are not combined into a general assessment of a bad position. Each requires a separate explanation and has its own scope.

3.2 Task Definitions and Scope of Analysis

The previous blocks examined how to resolve an existing check. In the new starting positions of this block, White's king **is not in check**. New functions and criteria are therefore defined rather than extending the earlier conclusion.

Two families of instructional positions are examined.

Family	Task under examination	Condition to preserve	Verification horizon
A. Restriction on a knight's action	Determine whether the c3 knight can capture the d5 pawn	The moving side's king must not be left in check	One White move from the specified position
B. Combining a capture with preservation of the queen	Capture the black pawn on d4	After the capture, Black must have no legal move that immediately captures White's queen	White's move and all immediate replies that capture the queen

The first condition is imposed by the rules of the game. The second is a **selected criterion of the analytical task**. These grounds must not be confused: the safety of one's own king constrains move legality, whereas preserving the queen may deliberately be excluded from the objective when examining a different combination.

Legality is determined by the FIDE Laws of Chess. A move must satisfy the applicable requirements for piece movement and must not leave the moving side's king in check or expose it to attack. ^[1]

This block does not identify the best move, evaluate the full material outcome of an exchange, or establish a win. The availability of a capture does not mean it is advantageous. The absence of an immediate capture of the queen does not establish its long-term safety.

3.3 Working Definitions

The definitions of state, structural relationship, function of an element, and availability of a function introduced in Block 2 remain in use. The following refinements apply to the analysis of several interacting functions.

Functional dependency

A **functional dependency** exists when the availability of a particular action, or the preservation of a required result, depends on the state of another element or on a particular relationship between elements.

Such a dependency must be expressed as a testable condition. For example, a knight's ability to move depends on whether the line to its own king remains blocked after it leaves its square.

Simply stating that pieces are connected is insufficient. The action under consideration, the condition to be preserved, and the evidence supporting the dependency must be specified.

Compatibility of functions

Compatibility of functions means that a legal action or continuation exists within the defined horizon that performs the required task while preserving the other specified conditions.

Compatibility relates to a particular state and set of requirements. It is not a permanent property of a piece.

If the conditions are met under one allocation of actions among pieces but not under another, the model must preserve that distinction.

Conflicting requirements for an element

A **conflict of requirements** arises when performing an action assigned to an element violates another condition that must be preserved.

The conflict may have different causes. The rules may prohibit the action. Alternatively, the action may be legal but fail to meet the adopted outcome criterion.

These cases must be recorded separately.

Alternative performing element

An **alternative performing element** is another element capable of performing the function under examination while meeting the same criteria.

The presence of another piece does not establish an alternative. A specific legal action must be presented and its result checked.

Dependency on a single element

Such a dependency is established when, in the specified state and horizon, a particular condition is maintained by only one element and a complete examination of the relevant possibilities finds no available substitute.

This conclusion cannot be reached simply by counting lines in a diagram. Other ways of changing the position must be considered, including moving the protected object or eliminating the attacking piece.

Several relationships involving one piece do not, by themselves, establish overload. A conflict requires a demonstration that specific requirements are incompatible.

3.4 Starting Positions

The new positions were constructed for this analysis. They are not attributed to historical games or presented as examples of optimal play.

Each position is reconstructed from the standard starting position. In the notation, **N** denotes a knight, **B** a bishop, and **Q** a queen.

Position A0. The only obstruction on the line to the king

1. d4 d5 2. Nc3 Nf6 3. e4 e6 4. e5 Bb4

White to move.

Relevant elements: White's king on e1, White's knight on c3, Black's bishop on b4, and Black's pawn on d5. The d2 square is empty.

No captures have occurred. All 32 pieces and pawns remain on the board.

Position A1. A control change to the attacking bishop's position

1. d4 d5 2. Nc3 Nf6 3. e4 e6 4. e5 Be7

White to move.

Compared with A0, Black's last move is replaced: the bishop is on e7 rather than b4. The material composition, the position of White's knight, and the position of the d5 pawn are preserved.

Position A2. An additional obstruction

1. d4 d5 2. Nc3 Nf6 3. e4 e6 4. e5 Bb4 5. Bd2 a6

White to move.

This position continues from A0. White's bishop has moved from c1 to d2, after which Black has played a7–a6. Black's bishop remains on b4.

Position B0. Capturing a pawn while the defender's movement is constrained

1. d4 d5 2. Nf3 Bg4 3. e3 e5 4. a3 exd4

White to move.

Relevant elements: White's queen on d1, White's knight on f3, White's pawn on e3, Black's bishop on g4, and Black's pawn on d4. The e2 square is empty.

The black pawn that originated on e7 has captured White's pawn on d4. The starting position B0 contains 31 pieces and pawns.

The complete sequences allow the reachability of the states to be checked, rather than merely the internal consistency of a chosen arrangement. This matters because the FIDE Laws of Chess define a position as illegal when it cannot be reached by any sequence of legal moves. ^[1]

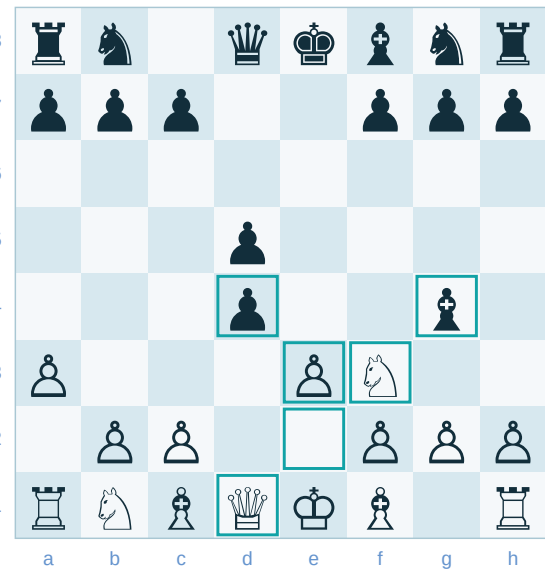
The diagrams correspond to the complete sequences in Section 3.4. White is to move in all four positions. Teal outlines mark squares relevant to the dependency under examination; every other piece is also retained on the board.

A1 · White to move



The bishop is on e7. The c3 knight has six legal moves.

B0 · White to move



The f3 knight blocks g4–d1 to White's queen.

3.5 Analysis of A0. An Attack Relationship Without an Available Capture

3.5.1. Two roles of one element

In position A0, the c3 knight participates in two relevant relationships.

First, it attacks the pawn on d5 in accordance with the knight's movement pattern. Second, it occupies an intermediate square on the line **b4–c3–d2–e1** between Black's bishop and White's king.

As long as the knight remains on c3, this line is blocked. White's king is not in check.

The move under examination, **Nc3xd5**, follows the knight's movement pattern and targets a square occupied by a black pawn. However, it vacates c3. Because d2 is also empty, the bishop's line of attack to the king on e1 opens.

Therefore, **Nxd5** is **illegal**. The reason is not an assessment of future losses, but the prohibition against exposing one's own king to attack. ^[1]

3.5.2. Checking the knight's other moves

From c3, the knight has six geometrical destinations not occupied by friendly elements:

a4, b1, b5, d5, e2, e4.

The other two geometrically reachable squares, a2 and d1, are occupied by White's own pawn and queen respectively.

Each of the six moves under consideration vacates c3. None creates a new obstruction on the line b4–c3–d2–e1 or eliminates Black's bishop.

Result: the c3 knight has no legal moves in this position.

This does not mean that White has no moves. For example, **5. a3** is legal. The restriction applies to a particular element in a particular state, not to the entire side's ability to continue the game.

3.5.3. Why an attack map is insufficient

Despite having no legal moves, the c3 knight is still considered to attack d5. Under the FIDE Laws of Chess, a piece is considered to attack a square even when moving to that square would expose its own king. ^[1]

Consequently, the statement that the knight attacks d5 cannot automatically be converted into the statement that the knight can capture on d5.

FINDING A0

One element participates in an attack relationship, but using that relationship to capture is prohibited because of another dependency involving king safety.

3.6 Control Checks for Family A

3.6.1. A1. Removing the specific cause of the restriction

In position A1, Black's bishop is on e7. The line b4–c3–d2–e1 is no longer a line of attack from that bishop. The move **5. Nxd5** is legal. White's king is not in check afterward.

Checking every knight move shows that the same six destinations considered in A0 now correspond to legal moves.

Thus, neither the knight's presence, its movement pattern, nor the target pawn's position has changed. What has changed is the specific dependency restricting the action.

This comparison concerns the selected function. Positions A0 and A1 are not treated as identical in other respects.

3.6.2. A2. Retaining the attacking piece and reallocating the function

In position A2, Black's bishop remains on b4, but a white bishop is now on d2.

After **6. Nxd5**, c3 is vacated, yet the bishop on d2 continues to block the line to the king. The knight's move is therefore legal.

The result is achieved not by eliminating the attacking piece, but by changing the allocation of functions: the knight can leave because another element maintains the required obstruction.

Comparing A0 with A2 is not a single-variable intervention. Two moves separate the positions, including a7–a6. The conclusion rests on a specific check: after the knight leaves, the d2 bishop continues to block the attack on e1.

3.6.3. Comparison of results

Parameter	A0	A1	A2
Black's bishop	b4	e7	b4
White's knight	c3	c3	c3
White bishop on d2	No	No	Yes
Knight attacks d5	Yes	Yes	Yes
Nc3xd5 is legal	No	Yes	Yes
Number of legal moves for the c3 knight	0	6	6

In A2, the dependency does not disappear entirely. Once the knight leaves, the d2 bishop maintains the obstruction on the specific line. The legality of its subsequent movement requires a new check.

Restoring one element's action is not the same as eliminating every dependency in the system.

Moreover, the legality of Nxd5 in A1 and A2 does not establish that the capture is advantageous. Black's pawn on e6 retains the option to recapture on d5. An overall evaluation of the exchange lies outside the current task.

3.7 Analysis of B0. A Legal Action That Fails the Joint Task

3.7.1. The joint criterion

In position B0, the task is to capture the black pawn on d4 without giving the opponent a legal immediate capture of White's queen on the next move.

The complete set of legal captures on d4 contains three actions:

5. Nxd4, 5. exd4, 5. Qxd4.

For each action, legality, the resulting arrangement, and the availability of a reply capturing the queen are checked separately.

3.7.2. Capturing with the knight

Before moving, the f3 knight blocks the line **g4–f3–e2–d1** between Black's bishop and White's queen.

After **5. Nxd4**, the knight occupies d4, the black pawn is removed, and f3 is vacated. The line to the queen opens.

Unlike A0, this move does not leave White's king in check. It is therefore legal. The rules do not prohibit a move solely because the opponent can capture the queen afterward. ^[1]

Black now has the legal reply:

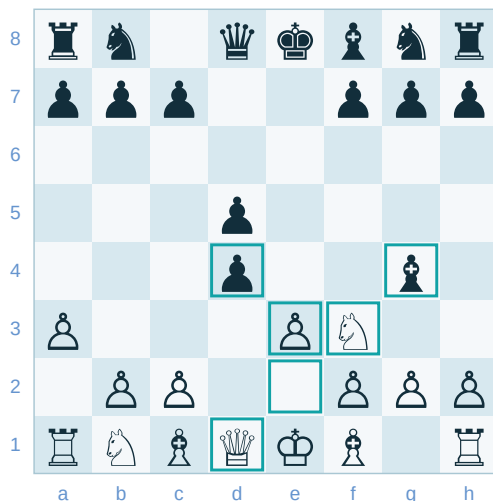
5... Bxd1.

The knight capture therefore fulfills the first part of the task but violates the second.

This does not mean that Black must capture the queen. The test concerns the availability of that legal reply. Nor does it determine the final evaluation of the entire combination after any subsequent captures.

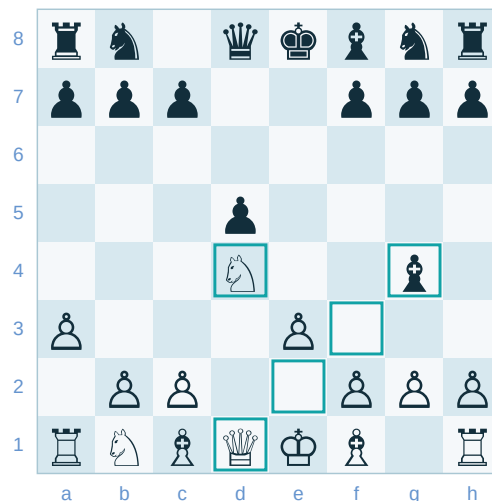
Result: the action is available under the rules but does not meet the specified joint criterion.

B0 · White to move



The f3 knight blocks g4–d1 to White's queen.

After 5. Nxd4 · Black to move



The f3 square is vacated. The reply 5... Bxd1 is available.

3.7.3. Capturing with the pawn

After **5. exd4**, White's pawn moves from e3 to d4 and removes the black pawn.

The knight remains on f3. The line g4–f3–e2–d1 remains blocked, and Bxd1 is illegal because the bishop cannot pass through the knight.

Checking the other immediate captures also finds no legal capture of White's queen on Black's next move.

The pawn performs the first function while the knight continues to maintain the obstruction. **The joint task is solved by allocating the functions to two elements.**

This is a specific example of an alternative performing element. The capability is established not by the mere presence of a pawn, but through a legal move and a check of the resulting state.

3.7.4. Capturing with the queen

After **5. Qxd4**, White's queen itself occupies d4 and removes the black pawn. The d2 and d3 squares are empty, so the path is available.

In the resulting position, Black has no legal immediate capture of White's queen.

Here, both the element performing the capture and the position of the protected element change. The previous question about access to the queen on d1 no longer describes its actual safety: the queen is now on a different square.

Preserving the result does not always require preserving the previous arrangement or defensive relationship.

This variation meets the adopted criterion over one opposing reply. Its long-term evaluation is not examined.

3.7.5. Results matrix

White's action	Move is legal	d4 pawn is captured	Black has an immediate capture of the queen	Joint criterion is met
5. Nxd4	Yes	Yes	Yes, Bxd1	No
5. exd4	Yes	Yes	No	Yes
5. Qxd4	Yes	Yes	No	Yes

After each of the three captures, 30 pieces and pawns remain on the board. The material composition immediately after White's move is the same, but the actions available to the opponent differ.

The matrix does not rank the variations by strength. It shows compliance with a specific criterion.

3.8 The Element Level and the Whole-Position Level

The results from the two families must be interpreted at the appropriate level.

In A0, there is no legal capture of the d5 pawn **by the c3 knight**. This is a conclusion about the action of a particular element. It does not mean that White has no other moves or cannot change the conditions for using the knight later.

In B0, the f3 knight cannot capture on d4 while also preserving the specified queen-safety condition. However, the joint task can be solved through other elements.

Level of conclusion	What is established	What does not follow
Action of an individual piece	Nc3xd5 is illegal in A0	White cannot continue the game at all
Assigning functions to one performing element	Nf3xd4 in B0 fails the joint criterion	The joint task has no solution
Capabilities of the entire side	B0 offers two suitable captures	Either capture is the best move
Change in a dependency	The d2 bishop allows the knight to leave in A2	The position is free of all defensive constraints

A restriction on an element and the absence of a system-level solution are different claims.

To move from the first claim to the second, alternative performing elements, other ways of performing the function, and legal changes to the protected state within the specified horizon must be excluded.

3.9 Verification Record

Preparation of the block included reconstruction of every starting sequence, checking the stated continuations, and an auxiliary programmatic enumeration of ordinary moves in the positions examined.

In the positions for which complete sets of actions are reported, castling, en passant, and promotion add no alternatives. Castling is blocked by the arrangement of the relevant pieces; the preceding moves create no en passant opportunity; no pawn can reach the last rank with a move under examination.

The auxiliary enumeration was used to check legality and immediate replies, not to assess playing strength.

Claim checked	Result
Every starting sequence can be reproduced with legal moves	Confirmed by internal checking
In A0, the c3 knight attacks d5 but has no legal moves	Confirmed
In A1 and A2, the c3 knight has six legal moves, including Nxd5	Confirmed
In B0, there are exactly three legal captures on d4	Confirmed
Bxd1 is available after Nxd4	Confirmed
No immediate capture of White's queen exists after exd4 or Qxd4	Confirmed

This status refers to internal reproduction and checking of specific claims. It does not denote independent review, certification of a software tool, or validation of the methodology over a broad set of positions.

To establish that a capability exists, a specific action is provided. To establish that it does not exist, the complete set of actions within the task's scope is examined.

3.10 Common Interpretation Errors

Error	Why the conclusion is invalid	Required clarification
A piece attacks a square, so it can capture there	The safety of its own king has not been considered	Check the legality of the resulting state
The move is legal, so the task is fulfilled	Additional outcome requirements have not been checked	Compare the result with every task condition
A piece has several functions, so it is overloaded	The number of relationships does not establish incompatibility	Demonstrate a specific conflict between actions
One performing element cannot do it, so no solution exists	Alternatives have not been checked	Examine the entire position
Another element has taken over the function, so the dependency is gone	The dependency may have shifted or changed	Re-establish the conditions for preserving the result
A safe immediate continuation was found, so the position is robust	The check has a limited horizon	Do not extend the conclusion beyond the replies examined

The distinction between an illegal action and an undesirable one is particularly important. In A0, the move violates the rules. In B0, the knight move violates only the selected joint criterion.

If the objective permits sacrificing the queen for another outcome, the task must be reformulated. The original criterion cannot be retained while its violation is simultaneously described as successful completion of the same task.

3.11 Acceptance Criteria and Deliverables

The block can be accepted after repeated reproduction if the starting states, stated action sets, and differences between results are confirmed.

A reviewer must obtain no legal moves for the c3 knight in A0 and confirm the legality of all six specified moves in A1 and A2. For B0, the three captures on d4 and the two variations meeting the joint criterion must be reproduced.

The consistency of conclusion wording is checked separately: prohibition under the rules must not be described merely as disadvantage, and a restriction on the selected performing element must not be described as impossibility for the entire side.

Deliverables of the block:

Deliverable	Content
Records of the new positions	Complete reconstruction sequences, side to move, and tasks under examination
Dependency descriptions	Relationship between the knight's action and blocking the line to the king or queen
Compatibility matrix	Comparison of three captures against two task conditions
Control-change record	Results for A0, A1, A2, and B0
Limitations register	Level of conclusion, verification horizon, and excluded generalizations

The formalization in Block 6 preserves the distinction between legality, performance of a function, and joint satisfaction of requirements. It also distinguishes the absence of a suitable action for a selected piece from the absence of a solution among the entire side's actions.

These distinctions are defined in domain terms and checked on the specified examples. The bounded mathematical specification is presented in Block 6.

3.12 Analytical Conclusion

Block 3 establishes three distinct mechanisms for restricting and restoring functions.

In A0, an attack relationship exists, but the corresponding capture is unavailable because it would expose the moving side's king. In A1, the capability becomes available after the attacking piece changes position. In A2, it is restored while the attacking piece remains in place because another element provides an additional obstruction.

In B0, the knight capture is legal but fails the joint task. Two other elements can perform the capture without an immediate loss of the queen within the defined horizon. A conflict of requirements for one performing element therefore does not become an absence of a solution for the entire side.

MAIN RESULT	A function must be assessed together with the dependencies that its action preserves or violates. An element's capability and the capability of the entire position must be checked separately.
-------------	---

This block is prepared for substantive review. The next block examines **a move as a transformation of the system**: action order, intermediate states, and how opposing replies affect the preservation of necessary capabilities.

Public Rules Reference

[1] FIDE. **Laws of Chess**. The rules governing move legality, attacked squares, and the reachability of positions. The reference is retained from the source text of this block.

<https://handbook.fide.com/chapter/E012023>

The analytical definitions and instructional positions belong to this demonstration analysis. The rules reference does not establish the methodology's universality or constitute an independent review of the findings.

STATUS	Version 0.2. Public working draft for review. Results are limited to the specified positions, tasks, and analytical horizon.
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Nelvarx

Structural analysis
of complex systems

STATE	ACTION
REPLY	CONDITIONS
ORDER	RESULT

STRUCTURAL ANALYSIS

Structural Analysis of Chess Positions

Block 4. A Move as a System Transformation

Move order, intermediate states, and opponent responses

DOCUMENT ID	NLV-CHESS-B04-EN
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AUTHOR	Alexey Gudilin
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Public demonstration material. This document presents a specific analysis; internal working materials are not included.

ALEXEY GUDILIN

Document Profile and Contents

Block 4 examines a move as a system transformation. It continues Blocks 1–3, moving from the functions and dependencies of individual elements to move order, intermediate states, and opponent replies.

RESULT	Preparation does not equal guaranteed execution. The order of one's own actions changes the possibilities for continuation and the opponent's replies.
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PUBLICATION BOUNDARY

Specific positions, findings, checks, and limits of applicability are presented. Internal work instructions and the full methodological toolkit are outside this document.

DRAFT STATUS

Version 0.2 is a Public working draft for review. Internal checks are reported to the specified extent; independent external verification is not claimed.

Position labels C and D correspond to B and F in the Russian edition.

4.1. Core Principle and Purpose of the Block

PRINCIPLE

The feasibility of a sequence of actions depends not only on its final objective, but also on the states that arise after each action and each opponent response.

The previous block examined the functions of elements and the dependencies that constrain their use. It established that an action by one piece may become available after another piece changes position, and that conflicting requirements assigned to one performer do not necessarily mean that the side as a whole has no solution.

Block 4 shifts the analysis from an individual action to a sequence of transformations. It examines move order, intermediate states, and the conditions under which the next action remains available after the opponent intervenes.

Central question: which conditions must remain in place at each stage for the plan to remain feasible?

The block is intended to produce checked descriptions of three mechanisms: creating an opportunity through a preparatory move, losing that opportunity through an opponent response, and the dependence of the outcome on the order of one's own actions.

It also examines a situation in which an immediate material gain does not prevent defeat. The local result and the outcome of the game are assessed separately.

4.2. Problem Statements and Scope of Analysis

The investigation uses two families of positions.

The first continues Example A0 from Block 3. The second introduces a separate instructional sequence involving the capture of a queen followed by a mating attack. The new continuations and their checks constitute the material of this block.

Family	Task under examination	Conditions to be checked	Horizon
C. Preparing an action	Create the opportunity for the knight on c3 to capture the pawn on d5, and execute the capture	Legality of each move, preservation of the performer and the necessary interposition	White's move, Black's reply, and White's next move
D. Order of actions	Force checkmate against the black king from a specified intermediate position	Legal transitions, a clear bishop route, coordination of the attacking pieces, and Black's replies	No more than two White moves, with a Black reply between them

In the description of the horizon, a move means one action by one player. A number in the game score denotes the conventional pair of White and Black moves.

Players move alternately. Two actions by the same side therefore cannot be treated as a continuous operation in which the state changes between stages only according to that side's plan. ^[4]

Family C does not assess whether capturing the pawn is advantageous. Family D does not establish the optimality of every move leading to its starting position. The examination of a forced result begins only from the explicitly identified state.

General opening evaluation, the statistical likelihood of errors, and the behavior of a particular opponent are outside the scope of this block.

4.3. Working Definitions

The definitions of state, function, dependency, and legal action introduced in the previous blocks remain in force.

State transition

A **state transition** is a change in the complete position resulting from one legal move.

The description records the starting state, the acting side, the move played, the resulting state, and the new side to move. Changes in relationships are determined from the resulting position, not merely from the identity of the piece that moved.

An illegal action does not constitute a transition in the game under analysis. The hypothetical arrangement after such an action may be used to explain why it is prohibited, but it is not included in the sequence of legal states.

Intermediate state

An **intermediate state** arises between the beginning of the plan under examination and the achievement of the target result.

Such a state must be legal and preserve the conditions required for continuation. An attractive intended final position does not compensate for the impossibility of passing through an intermediate stage.

Condition for the next action

A **condition for the next action** is a specific circumstance required to execute the planned move when the relevant side next has the turn.

The condition may concern the presence of the performer, a clear route, the position of the target, king safety, or the preservation of a particular interaction.

Conditions need not be identical at every stage. A square may first have to remain clear for a bishop to pass through it and later be used to place a knight.

Checked scenario

A **checked scenario** is a specific legal sequence of actions by both sides with an established result.

Its existence confirms that the stated course of play is possible. It does not establish that the opponent must choose the replies included in it.

Guaranteed result within a specified horizon

A result is **guaranteed within a specified horizon** when the side under examination has a way to achieve it against every legal opponent response within that horizon.

One successful branch is insufficient to support this conclusion. However, a single legal reply after which the required continuation is unavailable is sufficient to refute a specific claim of a guaranteed result.

This property is relative to the selected task and starting state. It does not replace an overall evaluation of the position.

4.4. Starting Positions

The notation uses **N** (knight), **B** (bishop), **Q** (queen), **K** (king), **+** (check), and **#** (checkmate).

Position C0. Preparing to move a pinned knight

The position is identical to A0 in Block 3:

1. d4 d5 2. Nc3 Nf6 3. e4 e6 4. e5 Bb4

White to move.

The white knight on c3 blocks the line **b4–c3–d2–e1** between the black bishop and the white king. The d2 square is empty. The knight attacks the pawn on d5, but an immediate **Nxd5** is illegal.

White's plan under examination is to place the bishop on d2 and subsequently capture the pawn on d5 with the knight from c3.

Position D0. Starting state before the pawn and queen captures

1. e4 e5 2. Nf3 d6 3. Bc4 Bg4 4. Nc3 Nc6

White to move.

Relevant elements: White's bishop on c4, knights on c3 and f3, and queen on d1; Black's king on e8, bishop on g4, knight on c6, and pawns on e5, d6, and f7.

All 32 pieces and pawns remain on the board.

Position D1. Starting point for checking the mating sequence

From D0, the following continuation is played:

5. Nxe5 Bxd1

White to move.

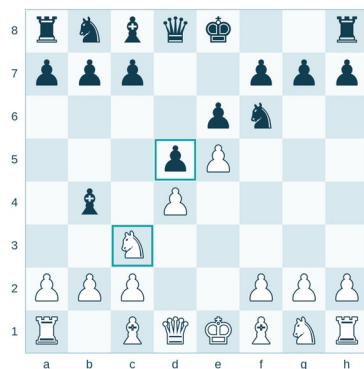
The white knight is on e5. The black bishop has captured the white queen on d1. White's bishop on c4 and knight on c3 remain in place.

There are 30 pieces and pawns on the board. The white king is not in check.

D1 is the starting state for comparing the two orders of White's actions. The transition to it is not treated as forced: Black could have selected a different fifth move.

All positions are reconstructed from the standard initial arrangement. Historical attribution of the sequences and claims of novelty for the chess combinations are outside the task.

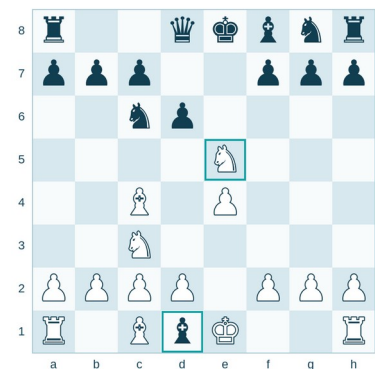
C0 · White to move



D0 · White to move



D1 · White to move



4.5. Family C. Preparation Does Not Guarantee Continuation

4.5.1. Why the immediate action is unavailable

In C0, **5. Nxd5** vacates c3 and opens the bishop's line from b4 to the king on e1. It is therefore illegal: a move may not expose one's own king to check. ^[1]

Placing the bishop on d2 after the knight has moved is insufficient. The required interposition must already exist in the state immediately resulting from the knight move.

The proposed sequence of first capturing the pawn and then restoring protection therefore has no legal first transition.

4.5.2. The preparatory move

White plays:

5. Bd2

An additional element appears on the line b4–c3–d2–e1. When the knight subsequently leaves, the occupied d2 square may continue to block the attack on the king.

However, it is Black's turn immediately after Bd2. The correct conclusion is therefore not that White can already execute Nxd5 immediately, but that **a condition for this action has been created, and its preservation depends on the opponent's reply.**

4.5.3. A reply that preserves the continuation

After:

5... a6

6. Nxd5 is legal. The knight leaves c3, but the bishop on d2 continues to block the line to the king.

This gives a legal sequence:

5. Bd2 a6 6. Nxd5

It confirms that the selected plan can be carried out against this particular Black reply.

The advantage of the capture is not established. In particular, Black's pawn on e6 retains the possibility of recapturing on d5.

4.5.4. A reply that removes the performer

Instead of a6, Black can play:

5... Bxc3

The bishop on b4 captures the knight on c3. The planned capture Nc3xd5 is no longer available because that knight has been removed from the board.

White could, for example, reply **6. Bxc3**, but this is a different action with a different objective. It does not execute the original plan of capturing the pawn on d5 with the knight.

State	What changed	Status of the planned knight capture
C0	The knight remains the only piece blocking the specified line	Illegal
After 5. Bd2	An additional interposition has been created; Black to move	Continuation depends on the reply
After 5... a6	The knight and the interposition remain	Nxd5 is legal
After 5... Bxc3	The knight has been captured	The action is unavailable

Family C result: the preparatory move creates the necessary opportunity, but does not guarantee preservation of every condition required to realize it.

This does not establish that White has no other plans. It refutes only the claim that, after Bd2, this knight is guaranteed to capture the pawn on White's next move.

4.6. Family D. The Outcome Depends on Move Order

Two White actions are considered in D1:

Bc4xf7 and **Nc3–d5**.

Both are legal as the first move from D1. Executing them in sequence, however, requires different states of the d5 square.

4.6.1. Letting the bishop pass first

White plays:

6. Bxf7+

The bishop travels along **c4–d5–e6–f7**, captures the pawn on f7, and checks the king on e8. The d5 and e6 squares are clear when the bishop passes through them.

Black has only one legal reply:

6... Ke7

White then plays:

7. Nd5#

The sequence is:

6. Bxf7+ Ke7 7. Nd5#

Initially, d5 serves as an intermediate square on the bishop's route. Once the bishop has passed through, it becomes the knight's destination.

4.6.2. Occupying the intermediate square first

Now, from the same D1 position, White starts with:

6. Nd5

The move is legal but does not give check. The knight occupies an intermediate square on the diagonal **c4–d5–e6–f7**.

The bishop cannot pass through a friendly piece. Placing the knight on d5 therefore obstructs the previously intended bishop route from c4 to f7. This follows from the rule that a bishop may not jump over pieces along its route. ^[1]

Black has the legal reply:

6... Nxe5

The knight on c6 captures the white knight on e5. In the resulting position, White's other knight remains on d5, and **Bc4xf7** is still illegal because the intermediate square is occupied.

The complete set of White moves was also checked: **none of the 33 legal moves delivers checkmate**.

Thus, **6. Nd5** does not guarantee checkmate by White's next move. **Nxe5** refutes that guarantee even if White is allowed to replace the originally planned continuation.

4.6.3. The essential dependency

The first order allows d5 to perform two different functions successively: first passage, then piece placement. The second order occupies the square needed for passage too early.

The distinction is not merely about the attractiveness of the final arrangement. It concerns **whether that arrangement can be reached through legal intermediate states**.

The same set of planned actions does not make their order interchangeable.

Reordering one's own moves also changes the opponent's legal replies. Two orders therefore cannot be compared while artificially retaining a Black reply that has become illegal in one of them.

4.7. Checking the Forced Finish in the Correct Order

4.7.1. Replies after 6. Bxf7+

After Bxf7+, the black king on e8 is attacked. Every legal reply must remove the check. The presence of check alone does not establish that the reply is unique; that requires examination of the full position. ^[4]

In this state, the check yields one reply: **Ke7**.

The d8 square is occupied by Black's queen, and f8 by Black's bishop. White's knight on e5 attacks d7 and f7. No other piece can capture the checking bishop, and there is no intermediate square between f7 and e8 on which to interpose.

The e7 square is available to the king. There are no other legal replies.

4.7.2. The state after 7. Nd5#

The knight on d5 attacks the king on e7. To verify checkmate, king escapes and captures of the checking knight must be ruled out.

King escape square	Reason it is unavailable
d6	Occupied by Black's own pawn
e6	Attacked by White's bishop on f7
f6	Attacked by White's knight on d5
d7	Attacked by White's knight on e5
f7	Occupied by White's bishop, protected by the knight on e5
d8	Occupied by Black's own queen
e8	Attacked by White's bishop on f7
f8	Occupied by Black's own bishop

There is no legal capture of the knight on d5. The knight's check cannot be blocked either, because its attack does not pass through intermediate squares.

The black king is in check and there are no legal replies. This constitutes checkmate and ends the game under the rules. Ordinary reply moves are not considered after this point. ^[4]

4.7.3. Exact scope of the conclusion

From D1, White can force checkmate by White's second move by choosing 6. Bxf7+.

This is stronger than a claim that one successful sequence exists: after the selected check, the complete set of replies, consisting of a single move, has been examined.

It does not mean that every White move from D1 produces the same result. The reversed order already gives Black a reply after which checkmate is unavailable within the specified horizon.

4.8. Local Gain and the Outcome of the Game

The transition **5... Bxd1** has an unambiguous immediate result: Black captures White's queen.

After that action, however, White still has the bishop on c4 and the knights on c3 and e5. Their interaction enables the checked mating sequence. Capturing the queen does not remove this mechanism.

Assessment point	Observed result	Supported conclusion
After 5... Bxd1	White's queen has been captured	Black has completed a specific capture
After 6. Bxf7+	Black's king is in check; only Ke7 is legal	The choice of reply is restricted to one move
After 7. Nd5#	Black has no legal reply	The game has ended in checkmate

Local gain is not a sufficient criterion for the success of a transformation. It must be compared with the target result and the other side's capabilities.

This is not a general claim that capturing a queen is harmful. The conclusion concerns this position and the checked continuation.

The boundary between a voluntary transition and a forced reply

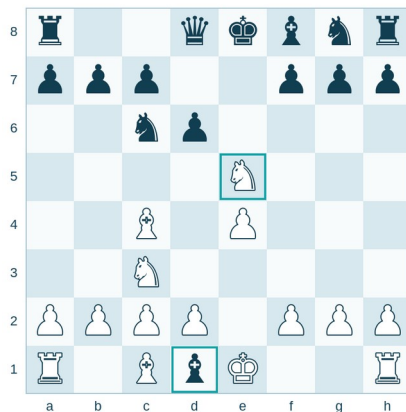
Black is not required to choose **5... Bxd1**. For example, after **5. Nxe5**, **5... Nxe5** is legal: the knight on c6 captures the white knight on e5.

If White then plays **Bxf7+**, Black has, among other options, **Nxf7**. The previous sequence with **Ke7** as the sole reply, followed by **Nd5#**, no longer applies.

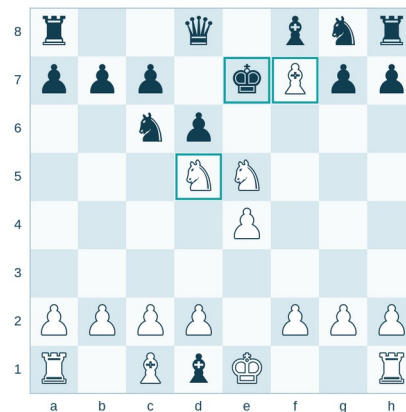
This does not establish a complete evaluation of the alternative branch. It shows why the proof of mate from D1 cannot be transferred to the earlier D0 position.

The analysis must distinguish **Black's choice to enter D1** from **Black's only legal reply after 6. Bxf7+**. The forced nature of the second does not make the first forced.

D1 · after 5... Bxd1



After 7. Nd5# · checkmate



4.9. Verification and Comparison Record

Preparation of this block included reconstruction of the starting sequences, checking of the stated transitions, and supplementary programmatic enumeration of legal actions.

A specific continuation is provided for a positive claim. The corresponding complete action sets were checked for claims of a unique reply, no replies, and no checkmate on the next move.

Claim checked	Internal check result
Nc3xd5 is illegal in C0	Confirmed
5. Bd2 a6 6. Nxd5 is a legal sequence	Confirmed
After 5. Bd2 Bxc3, the original knight on c3 is absent	Confirmed
In D1, both first moves, Bxf7+ and Nd5, are legal	Confirmed
After 6. Bxf7+, only Ke7 is legal	Confirmed
After 6. Bxf7+ Ke7 7. Nd5#, there are no legal replies	Confirmed
After 6. Nd5 Nxe5, Bc4xf7 is illegal	Confirmed
After 6. Nd5 Nxe5, no legal White move gives checkmate	Confirmed

The final check covers White's 33 legal moves and examines the state after each for checkmate. It is not a search for optimal play at greater depth.

The comparison of move orders from D1 produces the following result:

Parameter	Bxf7+ first	Nd5 first
First move is legal	Yes	Yes
Check is given	Yes	No
Number of legal Black replies	1	35
Relevant checked branch	Ke7, followed by Nd5#	Nxe5, after which there is no mate on the next move
Mate is guaranteed within two White moves	Yes	No

The number of replies alone is not an assessment of move quality. What matters is their content and their effect on the result under examination.

The check is internal. Independent reproduction remains a separate procedure before approval of the public version.

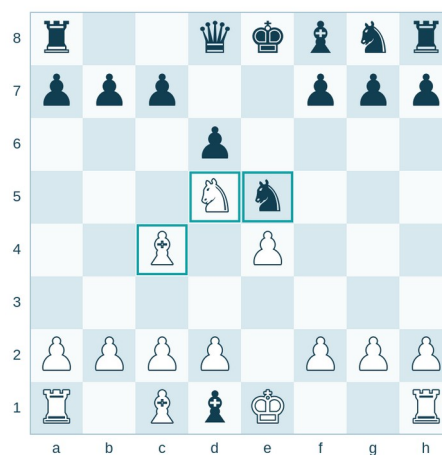
4.10. Common Errors in Sequence Analysis

Error	Consequence	Required correction
Evaluating the final arrangement without checking the route	An illegal intermediate state is included in the plan	Check every transition
Equating preparation with guaranteed execution	A reply that removes the performer or a required condition is ignored	Examine the state after the opponent intervenes
Considering one's own moves without the intervening reply	The actual sequence is replaced by actions of one side alone	Preserve alternating turns
Treating order as irrelevant	A friendly piece blocks the route needed for the next action	Specify each stage's conditions
Treating the capture of a valuable element as final success	An immediately attainable opponent result is overlooked	Continue the check to the adopted boundary
Extending a forced late reply to the whole sequence	Voluntary entry into a losing branch is presented as inevitable	Identify the exact starting state of the proof

Negative conclusions require particular care. The inability to execute the originally planned move does not by itself rule out another way to reach the objective.

Both checks were performed for the reversed order in Family D: Bxf7 was found to be illegal, and the absence of any mate on the next move after the selected reply Nxe5 was separately confirmed.

After 6. Nd5 Nxe5 · White to move



4.11. Acceptance Criteria and Deliverables

The block is accepted after repeat reproduction if the legality of the starting sequences, the stated restrictions, and the results of continuation checks are confirmed.

Consistency of states

The starting position and side to move must be unambiguous for every action. Analysis must not proceed from a hypothetical position without checking the transition leading to it.

The results for Family C must agree with A0 and A2 in Block 3. Reusing these states introduces no new starting assumptions.

Consistency of the task

The comparison of the two orders from D1 uses one criterion throughout: the ability to guarantee checkmate by White's second move.

It is not valid to check for mate in one order but consider only check or a captured piece in the other. A change in objective requires a separate problem statement.

Sufficiency of checking

For the correct order, the sole reply Ke7 and checkmate after Nd5 must be reproducible.

For the reversed order, the legality of Nxe5 and the absence of checkmate on White's next move must be confirmed. This is the branch that refutes a guaranteed result within the specified horizon.

Block deliverables:

Material	Contents
Starting-state records	Reconstruction sequences for C0, D0, and D1
Transition descriptions	Changes in performers, routes, interpositions, and the side to move
Move-order comparison	Results of two different first moves from the same position
Reply record	The sole reply in the mating branch and a counterexample for the reversed order
Limits of the conclusions	The starting states, criteria, and horizons to which the results apply

Block 6 formalizes the distinction between the existence of a successful scenario and a guaranteed result against legal replies.

Conditions must also be represented not only in the starting state, but before every subsequent action. The formalization must account for the fact that one's own move may either create a required relationship or prematurely remove a condition needed at the next stage.

4.12. Analytical Conclusion

Block 4 treats a move as a state transformation that changes not only an element's position, but also the possibilities for subsequent actions by both sides.

In Family C, the preparatory move creates an additional interposition, but the opponent may remove the performer of the next action. Creating a condition and guaranteeing execution of a plan are therefore different results.

In Family D, two legal first moves produce different possibilities for continuation. Letting the bishop pass before placing the knight forces mate from the stated position. The reversed order blocks a friendly piece's route and permits a reply that rules out mate within the same horizon.

The checked branch also establishes that capturing the queen does not prevent defeat. Immediate gain and achievement of the game's objective do not coincide.

RESULT

Main result of the block: a sequence must be assessed through the legality of its transitions, the preservation of conditions for the next action, and the consequences of opponent replies. Describing the final objective without these checks does not establish the feasibility of the plan.

The next block addresses **testing through changes in conditions**: systematic comparison of variants, identification of counterexamples, and refinement of the limits within which the conclusions apply.

Public Rules Reference

^[1] FIDE. Laws of Chess. The official chess rules cited in the source text of this block for alternating turns, move legality, and the definition of checkmate.

<https://handbook.fide.com/chapter/E012023>

Nelvarx

Structural analysis
of complex systems

CONDITIONS	CHANGES
TESTING	COUNTEREXAMPLE
BOUNDS	RESULT

STRUCTURAL ANALYSIS

Structural Analysis of Chess Positions

Block 5. Testing by Varying Conditions

Controlled changes, counterexamples, and limits of applicability

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Public demonstration material. This document presents a specific analysis; internal working materials are not included.

Document Profile and Contents

Block 5 continues Blocks 1–4: it brings the controlled checks together, expands the group of short games, and refines the limits of the conclusions. A separate pair of positions demonstrates the importance of move history.

RESULT	A tested conclusion includes the outcome, its conditions, the scope within which it is preserved, and known ways of changing it.
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PUBLICATION BOUNDARY

Specific positions, findings, checks, and limitations are presented. Internal work instructions and the complete methodological toolkit are outside this document.

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Position labels A, B, C, D, E correspond to А, Б, В, Г, Д in the Russian edition.

5.1. Core Principle and Purpose of the Block

PRINCIPLE

Testing must establish not only whether a result can be reproduced, but also which changes in conditions preserve it, alter it, or make it no longer applicable.

The preceding blocks examined the availability of defensive actions, dependencies between functions, and the effects of move order. Each conclusion was tied to a particular position and a bounded continuation.

Block 5 brings these results together in a system of controlled checks. Key examples are reproduced, the family of starting positions is expanded, and a new test of the completeness of the state description is introduced.

Central question: which changes actually affect the possibility under examination, and how far may the resulting conclusion legitimately be extended?

Testing is conducted at three levels. First, it establishes whether the previously described result can be reproduced. Next, it examines whether that result is preserved under specified changes. Finally, it tests claims that might arise from an unjustified generalization of the original example.

The deliverable is not an overall judgment that the methodology is correct or incorrect, but a **register of tested dependencies, counterexamples, and limits of applicability**.

5.2. Starting Basis and Problem Statements

The starting basis consists of the positions and definitions in Blocks 1–4. Their identifiers, functions, and criteria are retained. During retesting, the problem statement must not be changed in a way that automatically keeps the earlier conclusion valid.

An expanded group of short games, a counterexample with an alternative defense, and two positions with identical piece placement but different move histories are also introduced.

Test area	Question examined	Required result
Reproduction	Do the key results of the preceding blocks remain reproducible?	Confirmation or a recorded discrepancy
Controlled changes	Which selected changes preserve or alter the result under examination?	Comparison of conditions and available actions
Testing a generalization	Does a local conclusion hold beyond the original position?	Confirmation within new bounds or a counterexample
State completeness	Are piece placement and the side to move sufficient to determine available actions?	Identification of the additional information required

The FIDE Laws of Chess provide the rules basis. In particular, a legal move must not leave a player's own king in check or expose it to attack. The analytical model does not change this requirement; it uses it when checking states. ^[1]

The criterion and horizon are fixed separately for each task. For the short games, the state after the queen reaches h4 is examined. For functional dependencies, the tasks from Block 3 are retained. For move sequences, the horizon from Block 4 is retained. The new pair of positions concerns the availability of one specific capture.

The number of examples tested is not treated as independent evidence of universality. The set was constructed to test particular mechanisms, not as a random sample of all chess positions.

5.3. Unit of Testing and Conclusion Statuses

5.3.1. Unit of Testing

The unit of testing is a **specific claim about the availability of an action or the attainment of a result under stated conditions**.

The test record must include the starting position, the function under examination, the change introduced, the outcome criterion, and the boundary of the continuation being considered. The observed result and the grounds for the conclusion are then added.

For example, a claim that no defense exists after the queen reaches h4 must be linked to a particular move sequence. Without it, neither the king's location, the presence of obstacles, nor the possible replies has been established.

Testing whether an action exists and testing whether none exists require different scopes of work. For the former, one legal move is sufficient. For the latter, the entire set of actions capable of performing the stated function must be examined.

5.3.2. Controlled Change

A **controlled change** is an explicitly described difference from the original example, after which the same criterion is evaluated again.

When a move is replaced, all differences caused by that replacement must be considered. Replacing g4 with a3 changes the locations of two pawns relative to the final piece placement used for comparison. It must not be described as changing a single variable.

The comparison establishes the result in two states. To explain the mechanism, the analysis must also identify which action appeared or disappeared and which relationships caused the difference.

5.3.3. Statuses of a Tested Claim

The following statuses are used:

Confirmed within the stated bounds. The necessary scope of testing has been completed and supports the claim.

Refuted. A legal example or reply contradicting the claim has been found.

Not tested to the required extent. The available data or the continuations examined are insufficient.

Not applicable to this state. The condition that gives rise to the question is absent. For example, answering a current check is not required when the king is not in check.

These statuses apply to claims. They do not replace the function availability statuses introduced in Block 2.

A changed result does not necessarily mean that a test has failed. If a controlled change is expected to remove a defense and testing shows that the defense is absent, the expected difference is confirmed.

5.4. Extended Testing of the First Mechanism

5.4.1. Composition of the Test Group

Block 1 used three short sequences. For extended testing, a group of **16 variations** is formed, each starting from the standard initial position.

White's first move is selected from two alternatives: **f3 or f4**.

Black's first move is selected from two alternatives: **e5 or e6**.

White's second move is selected from four alternatives: **g4, a3, e4, or g3**.

Black's final move is **Qh4** in every case.

Every combination of these choices is tested: two first moves by White, two replies by Black, and four second moves by White. Arbitrary other continuations are outside this group.

All 32 pieces and pawns remain in every one of the 16 sequences. All the specified moves are legal.

The outcome under examination is assessed immediately after Qh4: whether White's king is in check and which legal replies exist.

5.4.2. Comparison Results

Within this group, replacing f3 with f4 and e5 with e6 does not change the classification below. All four combinations of first moves have been tested for each second move by White.

White's second move	Variations	Check after Qh4	Legal replies to check	Result
g4	4	Yes	None	Checkmate
a3	4	Yes	Only g2–g3	One defensive reply
e4	4	Yes	g2–g3 and Ke1–e2	Two defensive replies
g3	4	No	Answering check is not required	The line to the king is already blocked

The last row does not claim that no legal moves exist. On the contrary, play continues. It is specifically the question of answering a current check that becomes inapplicable.

After g3, the pawn occupies an intermediate square on the h4–g3–f2–e1 diagonal. The queen cannot attack the king through this pawn because sliding pieces cannot pass through obstacles. ^[1]

5.4.3. What Is Preserved and What Changes

Replacing f3 with f4 in this group preserves the condition relevant to the diagonal under examination: f2 has been vacated.

Replacing e5 with e6 preserves the queen's route: e7 has been vacated.

This does not mean that the corresponding pawn locations are equivalent in general. They may determine different attacks and other available moves. Only the selected classification has been shown to remain unchanged within the 16 tested variations.

White's second move changes the result in several ways. The move a3 retains the option of a later interposition. The move e4 also vacates e2 for the king. The move g3 establishes a block before the anticipated check can arise.

Result of the extension: the same material composition is compatible not only with the presence or absence of a defense, but also with different numbers of defensive actions or with no need to answer this particular check at all.

5.5. A Counterexample to Overgeneralization

5.5.1. The Broader Claim Being Tested

The first example might be incorrectly restated as follows:

If, after f2 is vacated, the g2 pawn advances to g4, the queen's move to h4 necessarily delivers checkmate.

This claim goes beyond the original conclusion. It accounts for the loss of g2–g3 but does not examine the other ways of defending.

A new sequence is introduced to test it:

1. f3 e5 2. e4 Nc6 3. g4 Qh4+

White to move.

The f2 pawn has moved, the g2 pawn is on g4, and the h4–g3–f2–e1 diagonal is open. The reply g2–g3 is unavailable. Nevertheless, this is not checkmate.

5.5.2. An Alternative Reply

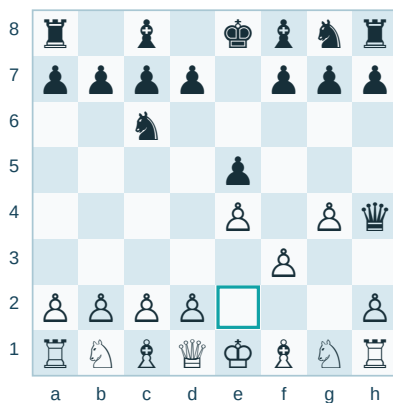
White's only legal reply is:

4. Ke2

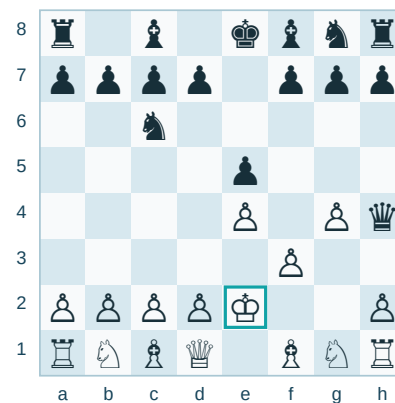
The move e2–e4 has vacated e2. In the resulting position, White's king is no longer in check.

The legality of this escape is determined by the safety of the destination square, not by whether another defensive method has been lost. The king may move to an adjacent square when the applicable constraints are satisfied. ^[1]

No captures have occurred before the queen's move. All 32 pieces and pawns remain on the board.



After 3... Qh4+: the only reply is 4. Ke2.



After 4. Ke2: check is answered. Black to move.

5.5.3. Significance of the Counterexample

The broader claim is refuted. Losing g2–g3 is not sufficient grounds for concluding that every position with the open diagonal is checkmate.

The original analysis remains correct: it checked separately that no king escape was available. The new example does not overturn that result; it demonstrates the need to retain the complete set of its conditions.

Losing one particular way of performing a function is not equivalent to losing the function until alternative ways have been ruled out.

This is consistent with Block 3, where a constraint on one performer was not equated with the absence of a solution for the side as a whole.

5.6. A New Test of State Completeness

5.6.1. Problem Statement

In Block 2, state was defined more broadly than piece placement. The new pair of examples tests this requirement directly.

The function under examination is to **capture the black pawn on d5 with the white pawn on e5 on the current move**.

Two sequences are compared.

Position E0:

1. e4 a6 2. e5 d5

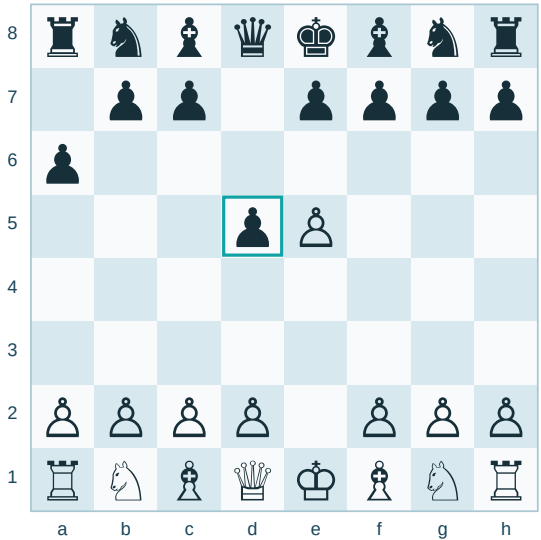
Position E1:

1. e4 d5 2. e5 a6

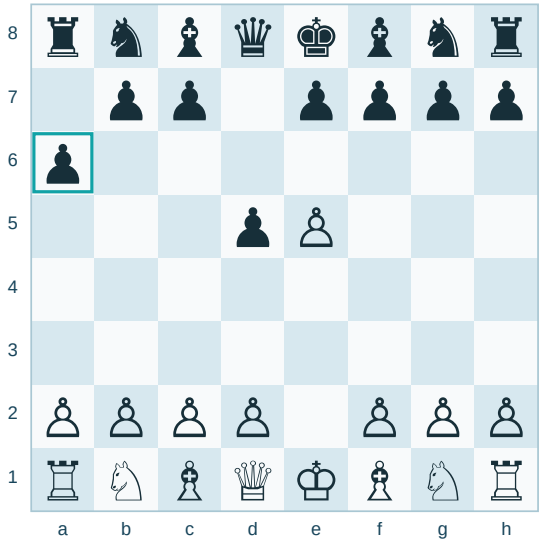
White is to move in both positions.

The piece placement is identical: the white pawn is on e5, the black pawns are on a6 and d5, and all other elements remain on their initial squares. Material composition and castling rights are also identical.

The order of Black's preceding moves is different.



E0. Last move: d7–d5. White to move.



E1. Last move: a7–a6. White to move.

5.6.2. Different Available Actions

In E0, Black's last action was d7–d5. White can therefore play:

3. exd6, en passant.

The white pawn moves from e5 to d6, and the black pawn is removed from d5.

In E1, the last move was a7–a6. The opportunity to capture the d5 pawn en passant is no longer available. This capture is permitted only immediately after the corresponding two-square pawn advance. ^[1]

Parameter	E0	E1
Complete piece placement	Identical to E1	Identical to E0
Side to move	White	White
Black's last move	d7–d5	a7–a6
e5xd6 en passant	Legal	Illegal
Total legal moves for White	31	30

Comparison of the complete move sets shows that they differ by exactly one action: **e5xd6 en passant**.

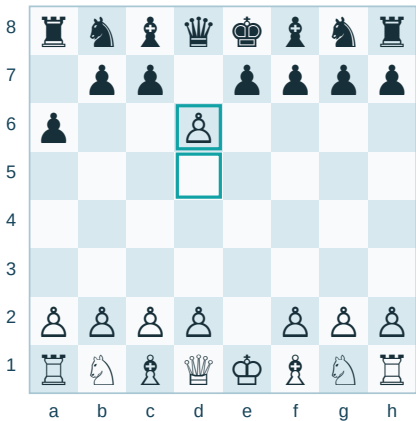
5.6.3. Test Result

Identical piece placement and the same side to move do not guarantee identical sets of legal actions.

These are not two identical complete positions in the sense of the model in Block 2. Only their spatial component is the same. A relevant right determined by the history differs.

If the analyst has only an image of the board without the necessary information about the previous move, the legality of e5xd6 cannot be reliably established. Missing information must be recorded as uncertainty, not as the absence of a right.

The new test confirms the need to include information that affects action availability in the state description. It does not require a change to the previously adopted definition, but demonstrates the error that arises when the state is reduced to piece placement alone.



E0 after 3. exd6: the white pawn is on d6, the black pawn on d5 is removed. Black to move.

5.7. Retesting Functional Dependencies and Sequences

To maintain consistency across the manual, key positions from Blocks 3 and 4 have been reconstructed again. The original sequences in Sections 3.4 and 4.4 were used without changing their criteria.

Test group	Result reproduced	Scope of the conclusion
A0, A1, A2	The knight on c3 has 0, 6, and 6 legal moves, respectively	The capabilities of that specific knight in the stated positions
B0	Three legal captures on d4 exist; two satisfy the combined criterion	Capturing the pawn without allowing an immediate capture of the queen in reply
C0 after Bd2	The reply a6 preserves Nxd5; Bxc3 removes the relevant knight	Preparation does not guarantee execution of the assigned action
D1 after Bxf7+	Only Ke7 is legal; Nd5 then gives mate	The result is forced within two White moves
D1 after Nd5 Nxe5	White has 33 legal moves; none delivers mate	This first move does not force mate within the same horizon

The Family A check retains the distinction between attacking a square and making a legal move. Under the FIDE Laws, a pinned piece may be considered to attack a square even though moving there would be illegal because of the safety of its own king. ^[1]

Checking the Horizon

D1 was also checked for an immediate checkmate on White's first move. No such moves exist.

However, the sequence **Bxf7+ Ke7 Nd5#** forces mate on White's second move. The claims that no mate in one exists and that mate can be forced within two moves therefore do not contradict each other. They concern different horizons.

After Nd5 as the first move, the reply Nxe5 rules out mate on White's next move. This does not prove the absence of mate at greater depth. Such a claim is not tested in this block.

Checking the Starting Boundary

The mating sequence starts from D1, after White's queen has already been captured. It does not automatically apply to the earlier position.

After **5. Nxe5**, Black may choose **5... Nxe5** instead of **5... Bxd1**. If White then plays Bxf7+, Nxf7 is available and the earlier sequence is no longer forced.

Changing the starting state or horizon changes the claim being tested, even when its wording appears similar.

5.8. Interpreting the Results of Changes

Preservation of the Selected Result

When a controlled change does not alter the result, preservation is established **for that comparison and that criterion**.

Within the group of 16 variations, pawn placements on f3 and f4 yield the same classification after Qh4 when the other selected moves are retained. This does not establish full equivalence between the corresponding positions.

A Change in Result

When the result changes, the analysis must identify which possibility has appeared or disappeared.

The e4 variation makes the king escape to e2 available. In the g3 variation, the line is already blocked. These cases prevent checkmate through different mechanisms and must not be treated as one mechanism merely because the final indicator is the same.

Refuting a Generalization

A counterexample refutes the claim it actually contradicts.

The sequence with Ke2 refutes the inevitability of mate after g2–g3 is lost. It does not refute checkmate in the original position from Block 1.

Identifying an Incomplete Description

Different outcomes under apparently identical conditions may mean that the conditions have not been fully described.

In E0 and E1, history explains the difference. There is no need to search for a change in piece placement that does not exist. The missing state attribute must be restored.

A conclusion that a condition is critical must remain tied to the task and the scope of testing. One change producing a different result does not establish that this condition is universally necessary for every possible way of reaching the objective.

5.9. Basis and Scope of the Testing Performed

The results in this block were obtained by reconstructing the sequences and using auxiliary software to enumerate legal actions.

For the expanded group, all 16 specified combinations were checked. For the new counterexample, the complete set of replies to check was established. For E0 and E1, the complete move sets were compared. For selected positions from the preceding blocks, the stated actions, replies, and final states were checked again.

En passant is accounted for explicitly in E0 and E1. Enumerating only ordinary pawn moves would be insufficient and would give the same result for two different states.

Checkmate is identified by checking both that the king is in check and that no legal moves exist. Having no legal moves without being in check is stalemate, not checkmate. ^[1]

This is an internal technical check of a specific set, not an independent assessment of the methodology. The descriptions of the examples and the auxiliary checking tool were prepared within the same work. Their agreement does not remove the need for independent reproduction.

No deep search for optimal play, statistical evaluation of analytical effectiveness, comparison with other methods, or tests on a random sample of games were performed.

The proportion of matching results in a deliberately selected set is not interpreted as the probability of correct performance on an arbitrary position.

5.10. Testing Risks and Controls

Risk	Possible distortion	Control
Selecting only confirming examples	A local conclusion appears broader than its evidence supports	Include changes that preserve the result and changes that create an alternative
Changing the criterion after observing the result	Any outcome is declared successful	Retain the original criterion or formulate a new task
Reducing the complete state to piece placement	Actions that depend on history are missed	Include relevant rights and preceding moves
Treating failure to find a reply as proof that none exists	An incomplete check becomes a categorical conclusion	State the actual extent of the actions examined
Confusing a scenario with a forced result	A convenient opponent reply is treated as obligatory	Test alternatives within the specified horizon
Treating one tool as independent confirmation	An implementation error may go unnoticed	Repeat the check by another means before final approval

Any discrepancy found must remain in the record. After the description is clarified or the check is corrected, the related examples must be reproduced again.

An inconvenient variation must not be removed from the set and the remaining results then presented as confirmation of the original, broader claim.

5.11. Acceptance Criteria and Deliverables

Reproducibility

All new positions must be reconstructible through the stated sequences from the standard starting position.

The group of 16 variations must reproduce four outcomes: checkmate, one defensive reply, two defensive replies, and no check. The initial material composition must remain unchanged in every variation.

For the new counterexample, **Ke2** must be confirmed as the only legal reply.

For E0 and E1, piece placement and the side to move must match, while the legal action sets must differ only by **e5xd6 en passant**.

Consistency of Conclusions

The results must preserve the distinctions between an element's action and the capabilities of the whole side, between no defense and no check, and between identical piece placement and identical complete states.

Confirmation of a bounded result must not become a claim of complete positional equivalence or universality of the method.

Deliverables

Deliverable	Contents
Register of tested claims	Criterion, starting state, horizon, and status
Matrix of 16 variations	Comparison of the selected moves and the availability of a defense
Counterexample record	An alternative king escape when g2–g3 is unavailable
E0/E1 pair	Testing how the legality of an action depends on history
Reproduction summary	Key results from Blocks 3–4
Register of limitations	Untested generalizations, depths, and application domains

The domain-specific dependencies tested here provide the basis for the bounded formalization in Block 6. Independent external verification is not claimed.

5.12. Analytical Conclusion

Block 5 extends testing from individual successful examples to comparisons of predefined changes.

The group of 16 sequences shows that the same material composition is compatible with no defense, one or two defensive replies, and no check at all. Changes that leave the selected classification unchanged do not make the complete positions equivalent.

The new counterexample establishes a boundary of the first mechanism: losing a particular interposing move does not imply checkmate if another legal way of answering check remains available.

E0 and E1 demonstrate another dependency: identical piece placement and the same side to move can coexist with different legal actions. An accurate description requires the relevant history.

Repeated checks of the key positions reproduce the results of the preceding blocks within their original bounds. Stronger claims about other horizons and arbitrary positions are not automatically confirmed.

Core result of the block: a tested conclusion must include not only the outcome, but also the conditions under which it was obtained, the scope within which it is preserved, and the known ways of changing it.

Block 6, Mathematical Formalization, presents states, legal actions, functional conditions, and bounded outcomes as a consistent specification that reproduces the examples tested here.

Rules Reference

^[1] FIDE. *Laws of Chess, effective 1 January 2023*.

FIDE Handbook: Laws of Chess

The source establishes the rules of play. Analytical findings apply only to the positions and testing boundaries explicitly stated in this block.

Nelvarx

Structural analysis
of complex systems

STATE	ACTIONS
TRANSITIONS	FUNCTIONS
HORIZON	VERIFICATION

STRUCTURAL ANALYSIS

Structural Analysis of Chess Positions

Block 6. Mathematical Formalization

States, actions, functional conditions, and bounded continuations

DOCUMENT	NLV-CHESS-B06-EN
VERSION	0.2
AUTHOR	Alexey Gudilin
ISSUE DATE	22 September 2026
STATUS	Public working draft for review

Public demonstration material. A bounded mathematical specification is presented; internal working materials are not included.

Document Profile and Contents

Block 6 continues Blocks 1–5 by introducing formal descriptions of states, actions, functions, and bounded continuations. The model is compared with the chess examples examined earlier.

RESULT	A bounded mathematical specification preserves the distinctions established in the domain analysis and reproduces the results of the demonstration set.	
6.1.	Core Principle and Purpose of the Block	66
6.2.	Scope of Application and Status of Statements	66
6.3.	State Representation	67
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PUBLICATION BOUNDARY

Notation, formal conditions, computed results, and the limitations of this specific case are presented. Internal work instructions and the complete methodological toolkit are outside this document.

DRAFT STATUS

Version 0.2 is a Public working draft for review. The formal clarifications recorded in Section 7.6 are incorporated here; independent external verification is not claimed.

Position labels A, B, C, D, E correspond to А, Б, В, Г, Д in the Russian edition. In the mathematical formulas, the horizon is measured in plies.

6.1. Core Principle and Purpose of the Block

The mathematical model must preserve the distinctions on which the domain analysis depends: between a state and its description, a legal action and task fulfillment, a successful scenario and an outcome that can be guaranteed against the opponent's replies.

The preceding blocks examined these distinctions in specific positions. Dependencies were established between material composition, function availability, the allocation of actions, the sequence of transitions, and move history.

Block 6 introduces formal notation and conditions that represent the results within a single model. Formalization does not change the original tasks or extend the scope of the conclusions already established.

Central question: can the mechanisms examined be expressed unambiguously, with computed results that match the positions analyzed earlier?

The deliverable is a bounded mathematical specification. It includes representations of states, legal actions and transitions, conditions for performing functions, checks of combined requirements, and a criterion for reaching a result within a specified horizon.

This is a formalization of the demonstration chess case. It is not a mathematical description of all Nelvarx activities or a model of optimal chess play.

6.2. Scope of Application and Status of Statements

The model covers standard chess, the positions from Blocks 1–5, and explicitly specified continuations. Action legality and chess outcome conditions are determined by the FIDE Laws in the edition used in the preceding blocks. The author's analytical definitions do not replace the rules of play.^[1]

Three types of statements must be distinguished.

A **definition** establishes the meaning of a symbol or condition. It must be consistent and appropriate to the task.

A **mathematical consequence** follows from the definitions and assumptions introduced. Its basis is a logical proof.

A **computed result** concerns a particular position or finite set of checks. It is supported by reproducing the corresponding actions.

These levels must remain consistent. A general claim cannot be declared proven merely because it agrees with several examples.

The model does not include clock time, arbiter actions, players' psychological characteristics, or estimates of error probability. Applying it to positions where draw claims, repetitions, or other grounds for ending the game matter requires those conditions to be handled accordingly. They do not determine the reported results in the short continuations examined here.

A bounded horizon does not permit a relevant rule to be ignored. If an unrepresented condition could change the answer, the task lies outside the model's current scope of application.

6.3. State Representation

6.3.1. Components of a State

Introduce the notation:

$$s = (b, \tau, c, e, \eta).$$

Symbol	Meaning
b	Complete placement of pieces and pawns
τ	Side to move
c	Remaining castling rights
e	Potential en passant target square, or the absence of such a square
η	History needed to reconstruct the state and check applicable conditions

The placement b maps each square to either an empty value or the element occupying it, together with its side and type.

If the task concerns a specific piece, such as the knight originally on c3, that element is assigned a persistent identifier. Movement changes its square but not its identity. This distinguishes use of the same acting element from replacement by another.

The representation is deliberately not claimed to be minimal. A complete move record may contain information already reflected in c and e . This redundancy allows the state to be checked against its history.

Castling rights and a potential en passant capture do not, by themselves, make the corresponding move legal: other legality conditions still apply.^[1]

6.3.2. Set of States

Let $\mathcal{S}$ denote the set of states admitted in the analysis.

For every initial state used, a legal reconstruction sequence is provided. An arbitrary arrangement is not included in $\mathcal{S}$ merely because it looks possible.

If relevant history is unknown, the analyst has an incomplete state description. A computation requiring missing information must be assigned the status **undetermined**, rather than performed by silently substituting a value.

This requirement agrees with the E0/E1 pair: identical piece placement and the same side to move were accompanied by different en passant possibilities.^[2]

6.4. Relations, Actions, and Transitions

6.4.1. Structural Relations

Let

$$\mathcal{R}(s)$$

denote the collection of relations defined in state s : square occupancy, attacks, interpositions, obstacles, and dependencies relevant to the task.

Relations must be derived from the state and the rules. They are not independent attributes that may be changed separately from the position.

For example, a record describing an open diagonal must agree with the actual occupancy of all its intermediate squares.

Attacking a square and being able to move there legally remain different relations. A piece may be considered to attack a square even though moving there is illegal because of the safety of its own king.^[1]

6.4.2. Legal Actions

Let

$$\mathcal{L}(s)$$

denote the set of moves satisfying movement, square-occupancy, special-rule, and king-safety requirements in the position.

A move is specified in sufficient detail to determine its origin square, destination square, and, where applicable, promotion. Its actual effect is established with reference to the state.

To exclude continuations after the game has ended, introduce the indicator:

$$\text{End}(s).$$

The set of actions with which the sequence under study may be continued is defined as

$$\mathcal{A}(s) = \begin{cases} \mathcal{L}(s), & \text{End}(s) = 0, \\ \emptyset, & \text{End}(s) = 1. \end{cases}$$

This distinction is necessary: the absence of a continuation after an established ending must not automatically be interpreted as stalemate or checkmate. The reason for termination is determined separately.

6.4.3. State Transition

For a legal continuation, define the function

$$T(s, a) = s', \quad a \in \mathcal{A}(s).$$

It returns the complete state after action a .

For given s and a , the transition result is unique. The choice of the action itself, however, belongs to the player and is not assumed to be predetermined.

If $a \notin \mathcal{A}(s)$, the transition is undefined within the analyzed sequence. An illegal action cannot be included in a plan as a temporary step to be corrected later.

A transition accounts for more than the movement of the acting element. A capture removes another element, castling moves two pieces, and an en passant capture removes a pawn from a square other than the destination. These effects must be reflected in the resulting state.

The outcome is checked using $T(s, a)$, not the visual appearance of the move notation.

6.5. Formal Description of the Function Under Study

6.5.1. Task Profile

The symbol Q denotes a specific analytical task. Its profile contains the side under consideration, the initial state, the fulfillment criterion, the conditions that must be preserved, and the testing horizon.

Introduce an applicability indicator:

$$\chi_Q(s) \in \{0, 1\}.$$

A value of 1 means that the task is applicable in the state. A value of 0 means that the condition giving rise to the task is absent.

For the task of answering a current check, applicability is determined by the presence of that check. Once the check has been answered, the task is no longer required in its previous form.

6.5.2. Fulfillment Condition

Introduce a testable condition:

$$F_Q(s, a, s') \in \{0, 1\}.$$

It takes the value 1 when the action and resulting state satisfy all the task requirements.

If the criterion includes an immediate reply by the opponent, that check is incorporated into F_Q . Its depth must match the horizon specified in the task profile.

For an applicable task in a state where the side under consideration is to move, define:

$$\mathcal{U}_Q(s) = \{a \in \mathcal{A}(s) : F_Q(s, a, T(s, a)) = 1\}.$$

This is the **set of task-satisfying actions**, not the set of all legal moves.

Two different conclusions follow:

$$\mathcal{U}_Q(s) \neq \emptyset$$

means that at least one suitable action exists;

$$\mathcal{U}_Q(s) = \emptyset$$

means that no such action exists within the specified scope of testing.

6.5.3. Availability Statuses

The four statuses from Block 2 are retained.

Available: the task is applicable and an action in $\mathcal{U}_Q(s)$ has been found.

Unavailable: the task is applicable and a complete check confirms that $\mathcal{U}_Q(s)$ is empty.

Not required: $\chi_Q(s) = 0$.

Undetermined: the state description, criterion, or check is insufficient for a conclusion.

The last status characterizes the completeness of knowledge and testing, rather than a special chess property of the position.

If the opponent is to move, the future availability of one's own action is assessed after that reply. It must not be described as absent merely because the other side moves now.

6.5.4. Capability of a Specific Acting Element

For an element p , define:

$$\mathcal{U}_Q(s; p) = \{a \in \mathcal{U}_Q(s) : \text{Mover}(s, a) = p\}.$$

Because this is a subset of $\mathcal{U}_Q(s)$,

$$\mathcal{U}_Q(s; p) \subseteq \mathcal{U}_Q(s).$$

The absence of suitable actions for the selected element therefore **does not justify concluding** that no suitable actions exist for the side as a whole.

6.6. Joint Fulfillment of Requirements

6.6.1. Intersection of Conditions

Suppose two requirements concern the same state and the same first action, with aligned horizons.

The joint condition is defined as

$$F_{1\wedge 2} = F_1 \wedge F_2.$$

The corresponding set is:

$$\mathcal{U}_{1\wedge 2}(s) = \mathcal{U}_1(s) \cap \mathcal{U}_2(s).$$

Hence

$$\mathcal{U}_{1\wedge 2}(s) \subseteq \mathcal{U}_1(s).$$

Adding a mandatory requirement cannot enlarge the set of actions satisfying the same task under unchanged initial conditions.

This follows from the definition of intersection; it is not an empirical assumption.

6.6.2. Application to B0

The position from Block 3 is retained:

1. d4 d5 2. Nf3 Bg4 3. e3 e5 4. a3 exd4

The first task is to capture the black pawn on d4:

$$\mathcal{U}_{\text{capture}}(s_{B0}) = \{\text{Nf3xd4}, \text{e3xd4}, \text{Qd1xd4}\}.$$

The second requirement is that, after the capture, Black must have no legal immediate capture of White's queen.

Let q be the persistent identifier of White's queen, and let $\text{Cap}(s', b)$ be the set of elements removed by action b .

The requirement is written as

$$\text{Alive}(q, s') \wedge \forall b \in \mathcal{A}(s') : q \notin \text{Cap}(s', b).$$

The check concerns the queen q , not its original square d1. After Qxd4, it occupies another square but remains the same protected element.

For the joint task, the result is:

$$\mathcal{U}_{\text{joint}}(s_{B0}) = \{\text{e3xd4}, \text{Qd1xd4}\}.$$

The knight on f3 has no suitable action:

$$\mathcal{U}_{\text{joint}}(s_{B0}; p_{Nf3}) = \emptyset.$$

The overall set is nevertheless nonempty. This formally represents the result from Block 3: **conflicting requirements imposed on one acting element do not exclude a solution using other elements.**

6.7. Defense Against Check and Preservation of Material Composition

6.7.1. Check and the Absence of a Defense

Let $C_i(s)$ mean that side i 's king is in check.

For either side i , checkmate is defined with an explicit side-to-move condition:

$$\text{Mate}_i(s) = [\tau(s) = i] \wedge C_i(s) \wedge [\mathcal{L}(s) = \emptyset].$$

Stalemate is defined as:

$$\text{Stalemate}_i(s) = [\tau(s) = i] \wedge \neg C_i(s) \wedge [\mathcal{L}(s) = \emptyset].$$

If side i is not to move, both indicators are zero. The absence of legal moves alone does not distinguish checkmate from stalemate; check must also be tested. [1]

In the examined positions under check, every legal reply must answer that check. Thus, for the narrow task from Block 1:

$$\mathcal{U}_{\text{defense}}(s) = \mathcal{L}(s).$$

This equality cannot be extended to every task. For capturing a pawn while preserving the queen, the set of suitable actions is considerably narrower than the set of all legal moves.

6.7.2. Material Description

Let $m(s)$ be the vector of piece and pawn counts by side and type.

It describes material composition but does not include placement, history, or available actions.

For the two original examples:

1. f3 e5 2. g4 Qh4#

and

1. f3 e5 2. a3 Qh4+

we obtain:

$$m(s_{\text{mate}}) = m(s_{\text{defense}}),$$

but

$$|\mathcal{U}_{\text{defense}}(s_{\text{mate}})| = 0, \quad |\mathcal{U}_{\text{defense}}(s_{\text{defense}})| = 1.$$

All 32 elements remain in both states. In the second, the reply g2–g3 exists.

Consequently,

$$m(s) = m(t) \nRightarrow \mathcal{U}_Q(s) = \mathcal{U}_Q(t).$$

A specific pair of states establishes this claim. It does not require an assumption about the behavior of every possible chess position.

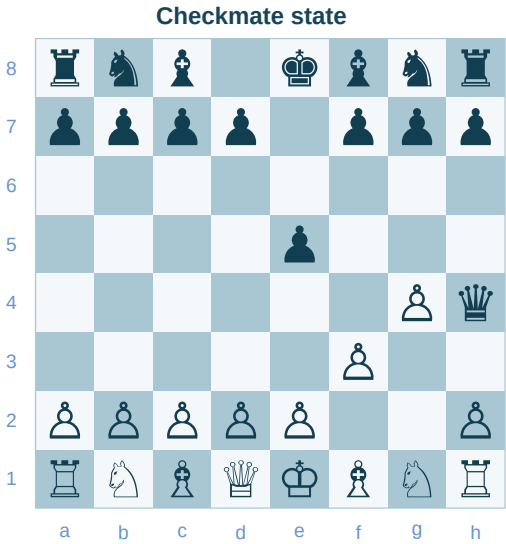
6.7.3. Extended Group

In the 16 variations from Block 5, substitutions of the opening moves preserved the following classification:

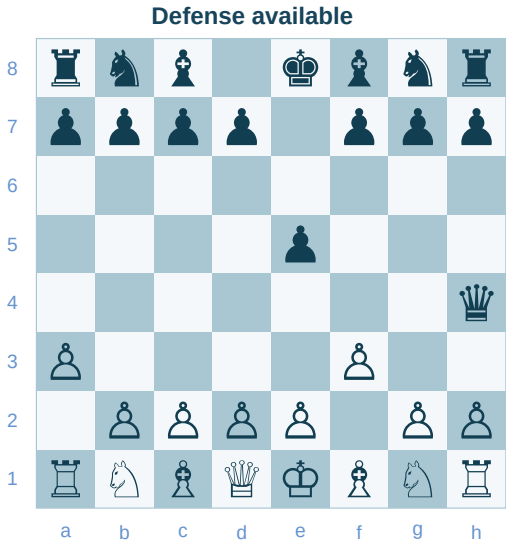
White's second move	Applicability of the task of answering check	Suitable replies
g4	Yes	None
a3	Yes	Only g3
e4	Yes	g3 and Ke2
g3	No	No current check needs to be answered

The final row is not assigned a numerical value of zero defenses. It represents a different task status.

Material composition is preserved in this set of sequences. It is **not declared an invariant of the entire game**, since other legal sequences include captures and promotions.



1. f3 e5 2. g4 Qh4#
No legal replies.



1. f3 e5 2. a3 Qh4+
The only reply is g3.

The states compared in Section 6.7.2. White to move in both positions; all 32 elements remain.

ALEXEY GUDIILIN

6.8. Sequences and Guaranteed Outcomes

6.8.1. Horizon

In the formulas, the horizon d is measured in consecutive player actions, or plies.

If the side under consideration moves first, $d = 3$ covers its first move, the opponent's reply, and its second move.

Thus, for the mating task in Block 4:

mate within two White moves means reaching the goal no later than the third ply from D1.

6.8.2. Scenario

The sequence

$$s_0 \xrightarrow{a_0} s_1 \xrightarrow{a_1} \dots \xrightarrow{a_{k-1}} s_k$$

is legal if, for every transition:

$$a_j \in \mathcal{A}(s_j), \quad s_{j+1} = T(s_j, a_j).$$

The existence of such a sequence with the required outcome establishes that a successful scenario exists. It does not yet establish that the result can be guaranteed against other replies by the opponent.

6.8.3. Strategy and Goal

For side i 's mating task, define the target condition:

$$G_i(s) = \text{Mate}_{\bar{i}}(s),$$

where $\bar{i}$ denotes the opposite side.

A strategy σ_i specifies a legal action for side i according to the state reached and the remaining horizon. It need not be a single predetermined sequence.

A guaranteed outcome means:

$$\exists \sigma_i \forall \sigma_{\bar{i}} : \quad G_i(s_k) = 1 \quad \text{for some } k \leq d.$$

The order of the conditions matters: one needs a strategy that works against every legal behavior of the opponent within the specified bounds.

6.8.4. Recursive Check

Let $W_d^i(s)$ indicate whether the goal can be guaranteed within at most d plies.

$$W_d^i(s) = \begin{cases} 1, & G_i(s) = 1, \\ 0, & G_i(s) = 0 \wedge (\text{End}(s) = 1 \vee d = 0), \\ \bigvee_{a \in \mathcal{A}(s)} W_{d-1}^i(T(s, a)), & \tau(s) = i, \\ \bigwedge_{a \in \mathcal{A}(s)} W_{d-1}^i(T(s, a)), & \tau(s) = \bar{i}. \end{cases}$$

The symbol $\bigvee$ means that one suitable move by the side itself is sufficient. The symbol $\bigwedge$ means that the result must remain achievable after every legal reply by the opponent.

The first two rows take precedence. A terminal state is evaluated before continuations are enumerated. This prevents an empty set of replies in a stalemate position from becoming a false confirmation of success through a universal condition over an empty set.

Correctness of the recursion relative to the specified finite tree follows by induction on depth. At zero horizon, the check is whether the goal has already been reached. At the next level, the side must be able to choose at least one suitable action, whereas all opponent replies must pass the check. Repeating this rule reproduces the definition of a guaranteed outcome.

6.8.5. Application to D1

The initial sequence is:

1. e4 e5 2. Nf3 d6 3. Bc4 Bg4 4. Nc3 Nc6 5. Nxe5 Bxd1

For the resulting state:

$$W_1^{\text{White}}(s_{D1}) = 0, \quad W_3^{\text{White}}(s_{D1}) = 1.$$

The positive value is supported by the strategy:

6. Bxf7+ Ke7 7. Nd5#

After Bxf7+, only Ke7 is legal. The next move ends the game in checkmate.

For a fixed first action with $d \geq 1$, $\tau(s) = i$, and $a \in \mathcal{A}(s)$, define:

$$W_d^i(s \mid a) = W_{d-1}^i(T(s, a)).$$

Then:

$$\begin{aligned} W_3^{\text{White}}(s_{D1} \mid \text{Bxf7+}) &= 1, \\ W_3^{\text{White}}(s_{D1} \mid \text{Nd5}) &= 0. \end{aligned}$$

In the second case, the legal reply **Nxe5** is sufficient. White then has 33 legal moves, none of which delivers mate.

The formal check therefore preserves the previously established distinction between the two move orders under the same horizon.^[2]

A zero value for the selected first move does not imply that no win exists at any depth. It concerns only the specified goal and horizon.

6.9. Changing Conditions and Incomplete Descriptions

6.9.1. Comparing States

A controlled change compares states s and s' , while task Q remains unchanged.

One can check whether task applicability, the set of suitable actions, or the guaranteed-outcome indicator is preserved.

Agreement on one indicator does not imply equality of states. For example,

$$|\mathcal{U}_Q(s)| = |\mathcal{U}_Q(s')|$$

does not establish that the actions themselves, their consequences, or the possibilities for further play are the same.

The record should therefore retain not only the final count but also the actions from which it was obtained.

6.9.2. Spatial Description as a Projection

Define the reduced description:

$$\pi(s) = (b, \tau, c).$$

It preserves piece placement, the side to move, and castling rights, but does not preserve the necessary en passant information.

For the examples from Block 5:

E0: 1. e4 a6 2. e5 d5

E1: 1. e4 d5 2. e5 a6

we have:

$$\pi(s_{E0}) = \pi(s_{E1}),$$

while:

$$|\mathcal{L}(s_{E0})| = 31, \quad |\mathcal{L}(s_{E1})| = 30.$$

The difference consists of one action:

$$\mathcal{L}(s_{E0}) \setminus \mathcal{L}(s_{E1}) = \{\text{e5xd6 en passant}\}.$$

These underlying results are recorded in Block 5.^[2]

6.9.3. Consequence for the Model

No function can return the correct legal-move set for both of these positions from the same value of $\pi(s)$ alone.

The proof follows directly from the pair: the input description is identical, whereas the required output sets differ.

The selected state reduction is therefore insufficient for this task.

This does not prohibit all reduced models. A reduction is acceptable if the omitted information is proven not to affect the result under consideration within the adopted scope.

6.10. Checking the Model Against the Examples

The positions and tested continuations were reproduced computationally again during preparation of this block. Auxiliary local checking code was used, rather than a chess engine's assessment of move strength.

Check	Formal representation of the result
16 variations of the first mechanism	Four classes of task applicability and defense availability were reproduced
Counterexample with e2 vacated	The only suitable reply is Ke2
A0, A1, A2	The knight on c3 has 0, 6, and 6 legal moves, respectively
B0	Three legal pawn captures; two satisfy the joint criterion
Preparation by Bd2	One reply preserves the continuation; another removes the acting element
D1	No mate on the first move; the chosen strategy forces mate within three plies
D1 after Nd5 Nxe5	All 33 legal moves were checked; no mate on the next move exists
E0 and E1	Identical piece placement; the move sets differ only by en passant

For the positive result in D1, a sufficient witness was checked: a move by the side, the complete set of opponent replies, and a mating continuation. There is no need to prove that this first move is the only way to reach the goal.

For the negative result after Nd5, an opponent reply was supplied and the complete set of White's next moves was checked. This excludes not only the originally planned continuation but also any mate within the remaining horizon.

The computations confirm that the formal conditions match the stated examples. They do not prove the auxiliary code correct for every chess situation and do not constitute an independent assessment.

This block does not claim a complete implementation of the specification, including extended game-ending conditions, as a finished standalone software product.

6.11. Acceptance Criteria and Deliverables

Consistency of Definitions

Each symbol must have one established meaning. The state, task, and check result must remain distinguishable.

The set of suitable actions must not contain an illegal move. If the task concerns a specific element, its identity must be preserved across transitions.

Consistency of the Horizon

The number of plies must agree with the verbal problem statement.

For a goal to be reached **no later than** a specified boundary, extending the horizon cannot remove a capability already established:

$$W_d^i(s) \leq W_{d+1}^i(s).$$

The reason is straightforward: a strategy that reaches the goal within d also reaches it within $d + 1$.

This property does not automatically apply to a different formulation, such as a requirement to reach the result on one exact move.

Consistency in Handling Uncertainty

Incomplete history or unfinished enumeration must not be encoded as a proven absence of an action.

A computed value of 0 is acceptable only after a check sufficient to support the corresponding negative claim.

Deliverables

Deliverable	Content
State specification	Placement, side to move, rights, and relevant history
Action and transition model	$\mathcal{L}$, $\mathcal{A}$, T , and stopping conditions
Functional model	Task applicability, fulfillment criterion, and the set of suitable actions
Joint-requirement model	Intersection of conditions and the distinction between an element's capabilities and those of the whole side
Bounded-continuation model	Goal, horizon, and recursive checking of a guaranteed outcome
Conformance record	Computed results for the examples from Blocks 1–5

The formal specification is presented and compared with the demonstration set. Its domains are clarified in Sections 6.7.1 and 6.8.5; Block 7 records the final alignment. Independent external verification is not claimed.

6.12. Analytical Conclusion

Block 6 represents the domain findings through states, relations, legal actions, transitions, and task-fulfillment conditions.

The formalization preserves the main distinctions from the preceding analyses. Equal material composition does not determine function availability. The absence of a suitable action for one element does not imply that the whole side has no solution. A move's legality does not guarantee fulfillment of combined requirements.

For sequences, a separate condition is introduced for guaranteeing the outcome against legal opponent replies. The E0/E1 pair shows why an insufficient state description cannot determine available actions unambiguously.

Core result of the block: a bounded mathematical model has been obtained that reproduces the mechanisms examined and explicitly states the data and checks required for each conclusion.

Block 7, Final Review and Assembly of the Demonstration Manual, aligns Blocks 1–6, records the confirmed results and remaining limitations, and states the status of this consolidated English edition.

Rules Reference and Series Materials

[1] FIDE. *Laws of Chess*, effective 1 January 2023.

[FIDE Handbook: Laws of Chess](#)

[2] Gudilin, A. Structural Analysis of Chess Positions. Block 5. Testing by Varying Conditions. In this consolidated edition, version 0.2, 2026. Sections 5.6–5.7 (pages 57–59).

The rules reference establishes the rules of play. Series materials record the underlying examples and internal checks. The formal specification in this block is limited to the explicitly stated tasks and horizons.

Nelvarx

Structural analysis
of complex systems

BLOCKS	RELATIONS
CHECKS	CLARIFICATIONS
BOUNDARIES	READINESS

STRUCTURAL ANALYSIS

Structural Analysis of Chess Positions

Block 7. Final Review and Assembly of the Demonstration Manual

Consistency of results, incorporated formula clarifications, and consolidated-edition status

DOCUMENT	NLV-CHESS-B07-EN
VERSION	0.2
AUTHOR	Alexey Gudilin
ISSUE DATE	22 September 2026
STATUS	Public working draft for review

Public demonstration material. The concluding block of consolidated English version 0.2; internal working materials are not included.

Document Profile and Contents

Block 7 completes Blocks 1–6: it compares results, records the formal clarifications incorporated into Block 6, and states the review status of this consolidated English manual.

RESULT	The consolidated English edition is prepared, with the Block 6 clarifications incorporated. Internal checks are recorded; independent external verification is not claimed.
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PUBLICATION BOUNDARY

Results, bases for conclusions, limitations, and acceptance conditions for the specific case are presented. Internal work instructions and the complete methodological toolkit are outside this document.

DRAFT STATUS

Version 0.2 is a Public working draft for review. The Section 7.6 clarifications are incorporated into Block 6. The English assembly is prepared; readiness of a revised Russian edition is not established here.

Position labels A, B, C, D, E correspond to А, Б, В, Г, Д in the Russian edition.

7.1. Core Principle and Purpose of the Block

An analytical case is complete when its problem statement, model, tested examples, and limits of the conclusions are consistent. Individual persuasive analyses do not replace a check of their compatibility.

Blocks 1–5 examined specific chess mechanisms and controlled changes to their conditions. Block 6 introduced a bounded mathematical specification that represents these findings through states, legal actions, functional requirements, and bounded continuations.

Block 7 completes the substantive structure of the manual. It compares results, records the clarifications incorporated into Block 6, and establishes the status of this consolidated English edition.

Central question: do the prepared materials form a single reproducible case in which the conclusions match the scope of the testing performed?

The deliverable is a final assessment of this consolidated English working draft, accompanied by a register of confirmed results, incorporated clarifications, and limitations.

Readiness of the demonstration manual does not mean that every possible study of chess has been completed or that the entire Nelvarx methodology has been tested in every domain. The object of acceptance is specific: the presented set of tasks, examples, and formal conditions.

7.2. Composition and Logic of the Completed Case

The material is organized into seven sequentially connected blocks.

Block	Task completed	Result used in the subsequent work
1. Problem Statement and First Example	Distinguish preservation of elements from availability of a defense	A reproducible mechanism for the loss of a defensive response
2. Structural Model of a Position	Connect the state, task, relations, functions, and actions	A common form of domain description
3. Functions and Dependencies of Elements	Test performer constraints and compatibility of requirements	A distinction between the capabilities of one piece and the whole side
4. A Move as a System Transformation	Examine move order and the effect of opponent replies	A distinction between a successful scenario and a guaranteed outcome
5. Testing by Varying Conditions	Extend the tests and establish the limits of generalization	A control group, a counterexample, and a test of relevant history
6. Mathematical Formalization	Express domain findings using consistent notation	A bounded specification of states, actions, and outcomes
7. Final Review and Assembly	Align the materials and define release conditions	A consolidated assessment and acceptance criteria

The connections between the blocks are substantive. The formal model must reproduce the results of the domain analyses, and the control examples must test the specific dependencies identified in those analyses.

Block 6 already establishes a boundary: agreement between calculations and individual examples does not prove that the checking tool is correct for every chess situation. This boundary remains in force in the final assessment. ^[2]

Assembly must not turn the seven blocks into a mechanically combined set of documents. The reading sequence, position labels, and contribution of each result to the overall task must be unambiguous to the reader.

7.3. Alignment of Terminology and Levels of Analysis

7.3.1. Required Distinctions

The consolidated edition preserves the following distinctions.

Concepts	Distinction to preserve
State and board arrangement	The state includes relevant rights and history, not only square occupancy
Attack and legal move	An attack relation does not establish that the corresponding move can be played
Legality and task fulfillment	A permitted move may fail to satisfy the chosen criterion
Element and function	The presence of a piece does not establish the availability of the required action
Performer and whole side	The absence of an action for one piece does not rule out an alternative
Scenario and guarantee	A successful branch does not imply success against every legal reply
Unavailability and uncertainty	An incomplete check does not establish that the action set is empty
Local result and game outcome	Capturing a piece does not replace checking the chosen final goal

These distinctions must be preserved in both the domain descriptions and the formulas. For example, the set of actions satisfying a task must not silently be replaced by the set of all legal moves. Block 6 gives these sets different notation and conditions. ^[2]

7.3.2. Alignment of Statuses

The statuses **available**, **unavailable**, **not required**, and **undetermined** are not interchangeable.

Once check has been resolved, the previous task is no longer required. If the history is unknown, the result may remain undetermined. If it is the opponent's turn, the side's future action is assessed after the opponent's reply, rather than declared absent. These conditions are preserved in the formal model. ^[2]

The verification status of a claim is recorded separately: it may be confirmed within the specified scope, refuted by a counterexample, or not yet tested to the required extent.

A function's status describes the capability under study. A verification status describes the basis for our conclusion about that capability.

7.4. Final Reproduction of the Test Set

7.4.1. Scope of the Testing Performed

A new internal programmatic reproduction of the key sequences was performed during preparation of this block.

Positions were reconstructed from the standard starting arrangement. The checks covered the legality of each transition, the sets of actions under examination, the presence of check, the absence of replies, and the outcomes of bounded continuations.

An auxiliary local checking program was used. It is not an independent expert and is not used to assess chess playing strength. Computation of a forced mate is restricted to the horizon defined in Block 6.

7.4.2. Comparison with the Recorded Results

Test group	Result of the current reproduction	Agreement with preceding blocks
16 short sequences	Four variations with checkmate, four with one reply, four with two replies, and four without check	Matches
Counterexample with e2 vacated	The only reply to check is Ke2	Matches
A0, A1, A2	The knight on c3 has 0, 6, and 6 legal moves, respectively	Matches
B0	Three legal captures on d4; two satisfy the combined criterion	Matches
Preparation with Bd2	The reply a6 preserves Nx d5; Bxc3 removes the designated performer	Matches
D1, correct order	After Bxf7+, only Ke7 is legal; Nd5 completes the sequence with checkmate	Matches
D1, reverse order	After Nd5, Black has 35 moves; after the selected Nxe5, White has 33 moves, none delivering mate	Matches
E0 and E1	31 and 30 legal moves; the only difference is the en passant capture e5xd6	Matches

Preservation of all 32 pieces and pawns was separately confirmed in the expanded group of 16 sequences. For E0 and E1, the action sets themselves were compared, not only their sizes.

No new discrepancies with the specified results were found within this reproduction. The check concerns the listed claims, not every possible continuation from each position.

7.4.3. Checking the Formal Horizon

For D1, the following values were obtained again:

$$W_1^{\text{White}}(s_{D1}) = 0, \quad W_3^{\text{White}}(s_{D1}) = 1.$$

There is no mate on White's first move from this position, but there is a way to force mate no later than the third ply.

With Bxf7+ fixed as the first move, the result is guaranteed. With Nd5 as the first move, the reply Nxe5 is available, after which no White action delivers checkmate within the remaining horizon.

These values agree with the results recorded in Section 6.8.5. ^[2]

The identified counter-reply rules out a guaranteed outcome within the stated bounds, but does not determine the course of the entire subsequent game.

7.5. Consolidated Register of Substantive Findings

Finding 1. Material Composition Does Not Determine Function Availability

Two positions can have the same composition of pieces and pawns but different defensive possibilities. This is established by comparing specific states, not by a general assumption about insufficient resources.

Practical use of the conclusion requires the function under study to be stated. Equality of composition alone does not establish equality of capabilities.

Finding 2. A Performer's Constraint Does Not Imply the Absence of a Solution

In A0, the knight's particular action is illegal because of king safety. In B0, the knight's action is legal but fails the combined criterion. Other elements nevertheless solve the task in B0.

The formal basis for the distinction is:

$$\mathcal{V}_Q(s; p) \subseteq \mathcal{V}_Q(s).$$

An empty subset does not establish that the entire set is empty. This relation was introduced in Block 6 and is consistent with the domain analysis. ^[2]

Finding 3. Action Order Is Part of the Mechanism

In D1, square d5 is initially required as an intermediate square on the bishop's route and later as the knight's destination. Occupying it prematurely changes both the side's own capabilities and the opponent's legal replies.

The assessment must therefore consider not only the desired arrangement, but also whether the path to it is legal.

Finding 4. A Local Explanation Requires Alternatives to Be Checked

Losing the g2–g3 reply does not imply checkmate if another legal way to resolve check remains. The Ke2 counterexample limits an overgeneralization, but does not invalidate the original analysis, in which the king's escape squares were checked separately. ^[3]

Finding 5. A Spatial Description May Be Insufficient

E0 and E1 have the same board arrangement and side to move, but differ in the availability of an en passant capture. The rule allows this capture only immediately after the relevant two-square pawn advance. ^[1]

This task therefore requires information that a single image of the board does not contain.

These findings form one domain-level conclusion: an action's availability is determined by the state, relations, task requirements, and legal transitions, rather than by an inventory of elements alone.

7.6. Clarifications to the Formal Notation for the Consolidated Edition

Version 0.2 incorporates two formal clarifications directly into Sections 6.7.1 and 6.8.5. This section records the changes already made to this consolidated English PDF; it does not describe pending edits to Block 6.

7.6.1. Checkmate and Stalemate Indicators for Either Side

Section 6.7.1 now explicitly includes the side-to-move condition in both indicators. Each is zero for a side that is not to move. This makes the recursive goal condition defined for either side throughout the state tree. [2]

The notation incorporated in Section 6.7.1 is:

$$\text{Mate}_i(s) = [\tau(s) = i] \wedge C_i(s) \wedge [\mathcal{L}(s) = \emptyset],$$

$$\text{Stalemate}_i(s) = [\tau(s) = i] \wedge \neg C_i(s) \wedge [\mathcal{L}(s) = \emptyset].$$

If side i is not to move, both indicators for that side take the value 0.

This makes the expression

$$G_i(s) = \text{Mate}_{\bar{i}}(s)$$

unambiguous in every state of the tree under examination. The results for the tested examples do not change.

7.6.2. Conditions for Fixing the First Action

Section 6.8.5 now states the conditions before the expression:

$$W_d^i(s \mid a) = W_{d-1}^i(T(s, a))$$

The incorporated conditions are:

$$d \geq 1, \quad \tau(s) = i, \quad a \in \mathcal{A}(s).$$

This means that at least one ply remains in the horizon, it is the side under consideration's turn, and the fixed action is legal.

Without these conditions, the expression cannot be interpreted as the result of choosing the side's first move. In particular, it does not apply to an illegal action or to a negative remaining horizon.

7.6.3. Preserving the Order of Terminal-State Checks

The recursive computation first checks whether the goal has been reached, then whether the game has ended or the horizon is exhausted, and only then examines continuations.

This order is retained unchanged in Section 6.8.4. Every implementation must preserve it: an empty set of replies cannot automatically count as success without identifying why the set is empty. [2]

The formula domains are now explicit in Block 6. The affected examples were rechecked during this revision; their move sequences and recorded results remain unchanged.

7.7. Evidence Status and Limits of the Final Review

7.7.1. What Is Confirmed

The results of the listed chess checks and their agreement with the bounded formal model are confirmed.

Mathematical consequences, such as inclusion of the set of actions satisfying combined requirements in the set satisfying one requirement, follow from definitions. Their basis differs from the basis of a result obtained by enumerating moves in a specific position.

The final material therefore preserves three categories: definition, logical consequence, and computed result.

7.7.2. What Has Not Been Established

The material does not establish superiority over other approaches, improvement in a user's chess play, an economic effect from business application, or reliability on an arbitrary position.

The software tool has not been comprehensively verified against every rule and ground for ending a game. Including the relevant notation in the specification is not equivalent to a complete software implementation.

The positions were selected to demonstrate particular mechanisms. Repeated checks of related variations are not independent observations from which a universal probability of success can be calculated.

These limitations continue the boundaries recorded in Blocks 5 and 6. [2], [3]

7.7.3. Independent Reproduction

The current check was performed within the same work. It is therefore described as internal.

The package under review does not contain a documented independent reproduction protocol listing the claims checked and the results. Feedback on writing quality and external assessment of individual passages do not replace such a protocol.

An independent check must reproduce the positions from the published input data, rather than merely agree with the conclusions already stated.

7.8. Structure of the Consolidated Manual

This consolidated English document contains an overall title and profile, a contents page for Blocks 1–7, the seven analytical blocks, and a change log.

The overall profile specifies the title, author, version, purpose, and scope. It states that the reader is presented with a demonstration of Nelvarx structural analysis applied to a bounded set of chess tasks.

The main body preserves the numbering of Sections 1.1–7.12. Repeated explanations may be replaced with internal references, but definitions and test conditions must not be shortened into ambiguous wording.

The block appendices retain notation, reconstruction sequences, and results. The overall change log is on page 91. Each position must remain reproducible: a diagram does not replace history where history affects legal actions.

Reused examples do not count as new tests merely because they appear in another block. For instance, C0 continues from state A0, and A2 is used in the analysis of action preparation.

The purpose of assembly is to preserve traceability, not to increase the page count.

7.9. Alignment of Language Editions and the Publication Boundary

7.9.1. Position Labels

The English materials already use the following consistent mapping:

Russian edition	English edition
A	A
Б	B
B	C
Г	D
Д	E

This mapping is retained in the consolidated package. In particular, Russian Г1 corresponds to English D1, and Russian Д0 to English E0. The mapping table is already stated in the English edition of Block 5. ^[3]

A reference to control variation A in Block 1 must include the block number. It must not be confused with position A0 in Block 3.

7.9.2. Substantive Equivalence

The translation must preserve criteria, horizons, statuses, and the strength of claims. Confirmed in this set must not become confirmed in general.

Matching page counts are not a criterion of equivalence. Meaning, move sequences, formulas, and table results are checked.

The design retains the name **Nelvarx** and the author’s real name. No separate emblem is introduced.

7.9.3. Publication Boundary

The public package includes initial positions, domain definitions, the published bounded model, the bases for conclusions, checks, and limitations.

Internal working templates, configurations, and materials that are not part of the published case are excluded. This boundary is already stated in the preceding documents. ^[3]

However, the basis for a specific published conclusion cannot be concealed behind a reference to a nonpublic methodology. The reader must be able to check the claimed result using the open materials.

7.10. Register of Remaining Risks

Risk	Potential consequence	Action before release
Implicit domain of a formula	Different implementations adopt different interpretations	Incorporated in 6.7.1 and 6.8.5; preserve these domains in later revisions
Mixed labels across languages	A different position is checked	Apply the mapping table and check cross-references
Reduced input data	The position cannot be reconstructed unambiguously	Preserve the sequences and relevant history
Counterexample omitted when shortening the text	A local result appears universal	Retain the counterexample and explain its significance
Diagram and notation disagree	The reader receives contradictory input data	Compare diagrams with the reconstructed positions
Overstated verification status	An internal calculation appears to be an independent expert review	Preserve the actual status of each confirmation

A correction affecting the state, criterion, or result requires the related examples to be retested. A purely presentational change requires checks of rendering and navigation, but must not automatically count as a new analytical test.

The change log must distinguish a content correction, a definition clarification, a translation, and a layout change.

7.11. Acceptance Criteria and Overall Readiness

7.11.1. Substantive Acceptance

Each main conclusion must be linked to a defined task, initial state, outcome criterion, and sufficient checking.

The material must retain counterexamples and limitations. Publishing only the confirming portions of the analyses while removing the conditions under which an explanation ceases to apply is not acceptable.

The formal model must reproduce the domain results without changing the original criterion.

7.11.2. Acceptance of the Consolidated Document

This consolidated English edition has been checked for completeness, formulas, diagrams, tables, contents, and internal navigation. Readiness or substantive equivalence of a revised Russian edition is not established by this English-only revision.

7.11.3. Readiness Matrix

Area	Current status
Domain analyses in Blocks 1–5	Prepared; reported internal checks retained.
Bounded formal model	Prepared; clarifications incorporated into 6.7.1 and 6.8.5.
Internal checks	Completed to the stated extent; affected examples rechecked for version 0.2.
Consolidated English PDF	Prepared; overall title, contents, folios, and navigation checked.
Independent external verification	Not claimed; no independent verification protocol is supplied.
Revised Russian consolidated edition	Readiness not established in this revision.
Publication status	Public working draft for review.
Final validation or certification	Not claimed. Further review is separate from publication as a working draft.

Decision at this stage: the English consolidated edition is prepared for publication as a Public working draft for review.

Version 0.2 records an English editorial consolidation and explicit formula domains. It does not present internal checks as independent external validation and makes no readiness claim for another language edition.

7.12. Final Analytical Conclusion

The demonstration case brings together problem statements, structural descriptions of positions, analysis of functions and dependencies, examination of sequences, controlled variations, and bounded mathematical formalization.

The repeated internal reproduction preserved the recorded results. The distinctions established between material composition, action availability, fulfillment of combined requirements, and guaranteed outcomes are consistent within the set studied.

The formal clarifications recorded in Section 7.6 are incorporated into Sections 6.7.1 and 6.8.5. The affected examples were rechecked without changing their sequences or recorded results.

The main result is not a new chess combination, but a complete demonstration analysis: a question is posed, a description is constructed, dependencies are identified, changes are tested, and the limits of the conclusion are established.

The manual demonstrates a specific application of Nelvarx structural analysis. It does not replace demonstration with a claim of universal applicability, nor present an internal check as external confirmation.

The consolidated English edition is prepared as version 0.2: Public working draft for review. Internal checks are completed to the stated extent; independent external verification is not claimed. Readiness of a revised Russian edition is not established here.

Normative Source and Series Materials

[1] FIDE. *Laws of Chess*. Effective from 1 January 2023.
[FIDE Handbook: Laws of Chess](#)

[2] Gudilin, A. Structural Analysis of Chess Positions. Block 6. Mathematical Formalization. In this consolidated edition, version 0.2, 2026. Sections 6.5–6.11 (pages 69–77).

[3] Gudilin, A. Structural Analysis of Chess Positions. Block 5. Testing by Varying Conditions. In this consolidated edition, version 0.2, 2026. Sections 5.6–5.9 (pages 57–60); document profile (page 52).

The normative source defines the rules of the game. Series materials record the initial examples, definitions, and results of internal checks. This block does not replace independent reproduction.

Change Log and Revision Checks

Version 0.2 | 22 September 2026

Base file: the 88-page English compilation, version 0.1. This release adds two front-matter pages and this change log. Source body pages 1–88 now appear on pages 3–90.

Location in this edition	Change made
6.7.1 (page 71)	The checkmate and stalemate indicators now include the side-to-move condition. Both are zero for a side that is not to move.
6.8.5 (page 74)	The fixed-first-action formula is preceded by its three conditions. The terminal-state checking order in 6.8.4 is unchanged.
7.6 and related status text	Clarifications are described as incorporated into Block 6, not pending. The record of the changes is retained.
7.11.3 and 7.12 (pages 89–90)	English assembly prepared; internal checks reported to their actual scope; independent external verification not claimed. Other-language readiness is not established.
1.11 and cross-block wording	Obsolete future-stage wording now points to the completed blocks. The Block 1 conclusion points to the bounded specification in Block 6.
Front matter and navigation	Overall title and seven-block contents added. Company: Nelvarx Systems AB. Main-text short name Nelvarx retained. Folios, contents, bookmarks, and references aligned.

REVISION CHECKS AND PRESERVATION

The affected checkmate/stalemate cases and the D1 fixed-first-action examples were reproduced in a bounded internal check. The explicit domains agree with the previously stated example results. Move sequences, diagrams, and result tables are retained; the recursion and its terminal-state order are unchanged.

All pages were rendered for layout review. Internal destinations, contents, formulas, and preserved diagram regions were checked. These checks support this edited English working draft; they do not constitute an independent external validation of the methodology or a general chess-software certification.

Public working draft for review

ALEXEY GUDILIN